

High-Power Laser Systems

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High-power laser sources are widely used in industrial precision processing and act as a new platform for strong-field physics research using peak power over petawatt. This review focuses on realizing high-energy solid-state disk and slab systems and the nonlinear-suppression strategies for high-power fiber systems using the functional fibers. First, the implementations and enabling technologies of the solid-state lasers for increasing peak power from gigawatt to petawatt are reviewed. Then the mechanisms and suppression strategies of the deterioration effects (including stimulated Raman scattering, stimulated Brillouin scattering, and transverse mode instability) in various fiber amplifiers are analyzed. At the same time, the mechanism and achievements of the current functional fibers are introduced. Finally, the challenges and perspectives of high-power solid-state and fiber amplifiers are summarized.

1. Introduction

High-power lasers are widely employed in fundamental and applied scientific research, national security, medical treatment, industrial processing, and other fields. The peak power of megawatt-class lasers is mainly used in material processing,^[1–5] including surface structuring^[6,7] and cleaving of glass.^[8–11] The peak power of gigawatt-class lasers produces isolated attosecond extreme-ultraviolet pulses using high harmonic generation. It is a key technology for analyzing ultrafast electronic phenomena in attosecond physics.^[12–16] The peak power of terawatt class lasers is applied to filament atmospheric research,^[17,18] particle acceleration,^[19–25] X-ray lasers,^[26] inverse Compton scattering,^[27,28] etc. The peak power of petawatt class lasers is generally used in high-energy-density plasma physics,^[29,30] materials and laboratory astrophysics,^[31–33] laser plasma-wake field acceleration,^[19,34–37] quantum electrodynamics,^[38–42] and nuclear photonics.^[20,43,44]

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The first laser invented by Maiman was a ruby rod laser pumped by a flashlamp.^[45] The development of high-quality pump sources has enabled more advanced laser configuration designs to be realized. Pressing rod-shaped crystals into thin slices of a slab (Figure 1a) or disk (Figure 1b) and drawing them into optical fibers (Figure 1c) significantly improve the heat dissipation performance of the laser.

On the one hand, the optical fiber has the most significant heat dissipation area, and a 500 kW power output of continuous wave (CW) has been reported.^[46] However, the fiber form of the gain medium reduces the energy storage volume and thus limits the laser's output.

Currently, the doped core diameter of the fiber is generally less than 100 μm , making large pulse energy output impractical and limiting the peak power to several GW with chirped pulse amplification (CPA).^[47] On the other hand, taking advantage of the excellent energy storage volume, the traditional slab laser (zigzag & active-mirror slab in Figure 1d) can reach ≈ 1 GW peak power with ≈ 10 ns pulse width ≈ 10 J energy.^[48,49] The integrated thin-film multislabs with a more substantial heat dissipation capability allows larger-sized crystals, which increase the energy of the 10 ns pulse to ≈ 100 J level with peak power over 10 GW.^[50,51]

With sub-mm thickness, the disk has a greater energy storage capacity than optical fiber and a higher heat dissipation area ratio than the slab, allowing it to produce CPA-free average power of kW level, picosecond (ps) pulsed output energy of mJ level, and peak power exceeding GW.^[52,53] To generate TW-level peak power output, a regenerative preamplifier reduces the repetition frequency followed by a multipass master amplifier. This hybrid disk amplification construction is shown in Figure 1e. One of the primary research topics in multipass disk amplifiers is removing the focus from the crystal and the transmission path. The current approach in Figure 1i is to realize collimated transmission.^[53,54] To further increase the peak power to the PW class, a high-energy (100 J class) pulsed laser was used as the pump of the CPA or optical parameter chirped pulse amplification (OPCPA) laser system.^[55,56]

Auxiliary heat dissipation methods such as pulsed diode laser (PDL) pumping,^[57–61] zero-phonon-line (ZPL) pumping,^[54,62,63] cryogenic cooling,^[61,64,65] or gas cooling^[50,66] are required in addition to the basic configuration's heat dissipation benefits. Furthermore, large-aperture crystals' amplified spontaneous emission (ASE) must be reduced by eliminating the edge oscillation conditions.^[67–70] Figure 1h depicts a common edging approach

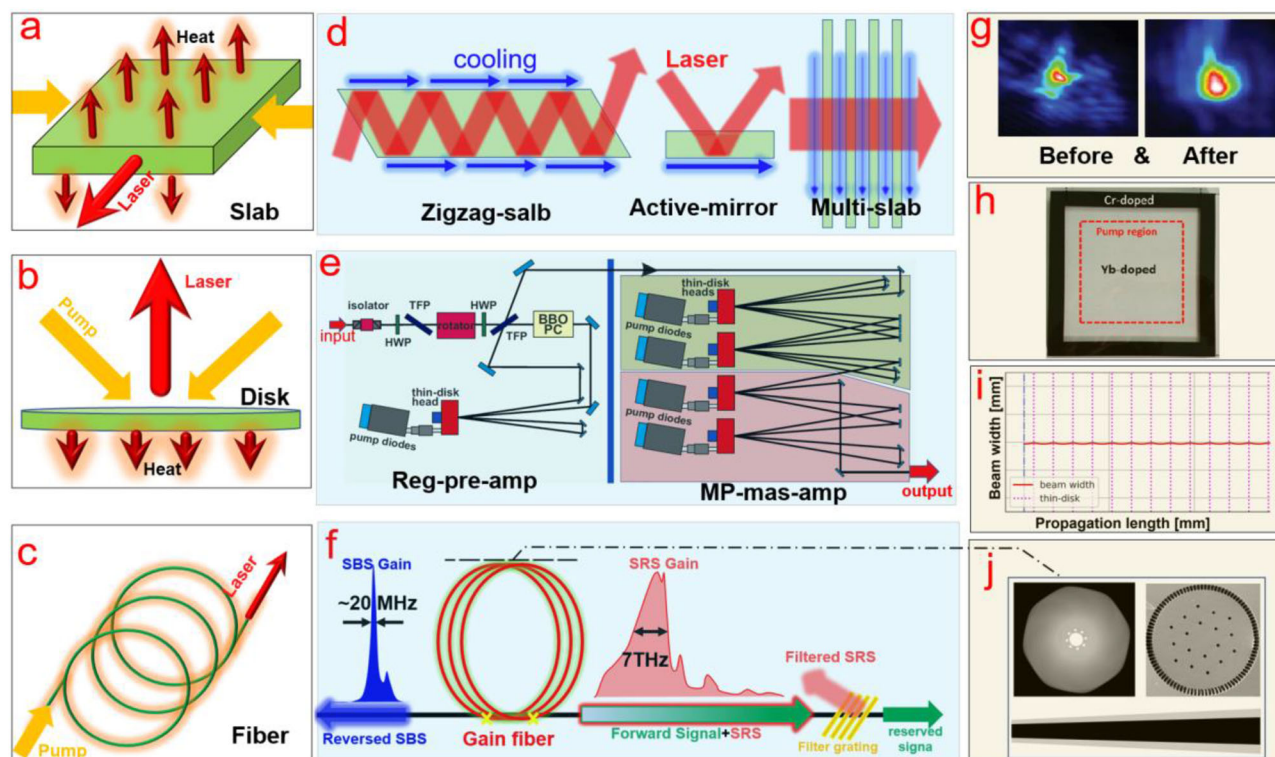


Figure 1. Schematic diagrams of high-power lasers and the related technologies. Concise schematic diagram of the structure of a) slab, b) disk, and c) fiber laser. d) Constructions of three typical slab laser (the blue arrows represent the cooling medium, and the thick red arrows represent the laser beam). e) Hybrid amplification system of disk: Reg-pre-amp (regenerative preamplifier) and MP-mas-amp (Multipass master-amplifier). f) Schematic diagram of the characteristics of Raman scattering and Brillouin scattering in fiber lasers: the blue and red spectral envelopes correspond to the gain spectrum of stimulated Brillouin & Raman scattering respectively, reverse blue arrows indicate the return light induced by stimulated Brillouin scattering, and the forward-propagating arrows (green filled with red-edged) represents the amplified laser signal with Raman noise. The Raman noise is filtered out by the tilted filter grating (represented by the yellow line array), and finally a pure signal (pure green filled) output is obtained. j) Cross-sectional schematic diagram of three typical special fibers. g) A beam spot energy distribution before and after correction. h) A 100 mm square slab surrounded by a 10 mm wide Cr-doped absorptive cladding for ASE suppression. i) Schematic plot of the beam width evolution during the propagation within the multi-pass disk cell. e) Reproduced with permission.^[81] Copyright 2020, Optical Society of America. g) Reproduced with permission.^[56] Copyright 2017, Optical Society of America. h) Reproduced with permission.^[71] Copyright 2015, Society of Photo-Optical Instrumentation Engineers (SPIE). i) Reproduced with permission.^[53] Copyright 2020, Optical Society of America. j) Reproduced with permission.^[79] Copyright 2015, Society of Photo-Optical Instrumentation Engineers (SPIE).

for suppressing edge oscillation.^[71] However, even if the most advanced thermal managements are used, it is difficult to completely eliminate the wavefront distortion in the high-intensity amplification process.^[72,73] The laser wavefront distortion caused by the thermal effect of the amplifier can be compensated to improve the laser beam quality by using phase conjugate mirror or adaptive optics.^[74,75] A beam spot energy distribution before and after correction is illustrated in Figure 1g.^[56]

For high-power fiber amplifiers, the key to achieving high peak power lies in the suppression of nonlinear effects such as stimulated Raman scattering (SRS), stimulated Brillouin scattering (SBS), and transverse mode instability (TMI).^[76] The inelastic scattering process can transfer the signal light to the Stokes light in the same direction or the reverse direction (as shown in Figure 1f). At present, in addition to the corresponding active suppression methods, one mainly designs special optical fibers to eliminate them while maintaining large-mode area (LMA) single-mode (SM) output. Some typical designs of fiber and filter gratings are shown in Figure 1j.^[77–80]

2. Solid-State Lasers with High-Peak-Power

2.1. From Sub-GW to 10 GW

Rod-type and zigzag slab amplifiers are the preferred solutions to achieve nanosecond pulse energy of 10 J class with peak power of 100 MW to 1 GW. Rod-type amplifiers are simple to pump and capable of realizing high energy storage. The laser beam does not pass through the coolant to avoid the influence of the coolant on the wavefront. At present, it is mainly used in high-energy pulsed lasers pumped by flash lamps^[82] and the preamplification stage of high-energy ns/ps lasers.^[72,83–85] There are also examples where high-energy pulsed lasers are directly realized by diode laser pumped rod lasers. Xu et al. reported that PDL-pumped Nd:YLF rods achieved an output of 9 J, 13 ns, 10 Hz, and 1053 nm, corresponding to a peak power of 692.3 MW.^[86] However, the radial heat flow of the rod amplifier will produce a significant transverse temperature gradient, which will cause optical distortion^[87] and is harmful to increase the pulse energy

further. The zigzag slab amplifier has a larger surface-area-to-volume ratio than traditional rod-type, and heat dissipation occurs in a direction transverse to beam propagation. Fan et al. used three-stage series zigzag Nd:YAG amplifiers and reported an output of 5 J, 6.6 ns, 200 Hz, and 1064 nm, corresponding to a peak power of 757.6 MW.^[72] The seed beam propagates down the slab by multiple total internal reflections, forming an unextracted energy storage area and reducing the amplification efficiency.^[87] Temperature gradients limit power scaling in the gain medium before the medium is damaged.

Compared with the rod and zigzag slab geometries, the active mirror configuration of the slab amplifier can achieve an output pulse with a peak power of more than 1 GW by balancing the thickness of the gain medium and heat dissipation. A large surface of the slab gain medium is used as a reflector. The laser beam is folded in a V-shaped optical path to extract the energy storage from the gain medium, and the reflective surface is also used as a pumping and cooling window for the gain medium. The advantages of this configuration are briefed as follows: 1) The beam passes through the gain module in a single pass to achieve double-pass extraction, with compact structure and high extraction efficiency; 2) The cooling channel is separated from the laser light path to avoid the influence of the fluid on the wave surface; 3) Pump light and V-shaped laser mode can realize high matching efficiency; 4) The thermal gradient is approximately along the optical axis. The disadvantage of this configuration is that there is only one large surface to dissipate heat. It is necessary to focus on eliminating the influence of the activation of the mirror deformation on the output under strong heat conditions.^[88] Both the premagnification and main amplification stage of the LUCIA laser system adopted a PDL active mirror configuration. Using a Φ 45 mm $\times$ 7 mm Yb:YAG ceramic, the laser system realized 13.9 J at 2 Hz output, the pulse width was 8 ns, and the peak power reached 1.7 GW.^[89] Many researchers applied multiple active mirror gain media chains to distribute heat accumulation and pump energy to each gain medium, achieving pulse energy greater than 10 J, repetition frequency 10 Hz, pulse width on the order of 10 ns, and peak power greater than 1 GW.^[48,83,90]

Another way to achieve pulses with GW peak power is CPA-free multipass thin-disk amplifiers, which directly produce ultrafast pulses with kW-level average power due to excellent heat dissipation performance.^[53,62,91,92] Without CPA or coherent beam combination (CBC), thin-disk multipass amplifiers can directly amplify the ultrafast seeding pulses at sub-MHz from tenths of μ J to mJ level and produce laser pulse with peak power up to GW.^[52,93]

The employment of a collimated seed beam thin-disk multipass amplifier, plane optics (plane mirrors), and a thin disk with a wide radius of curvature is ideally suited for scaling both the average and peak power for femtosecond (fs) or ps operation.^[2] The active medium has a diameter on the order of sub-mm and a typical thickness of 100–300 μ m. Due to the large spot diameter having weak nonlinear effects and a small thermal lens effect, the phase front distortion is minor.^[67,94,95] To overcome the small gain of one trip, the seed pulses must have good transmission characteristics to ensure multiple-pass for efficient energy extraction and match the pump spot.^[2,68,92,96–100] Based on the principle of collimated propagation, the number of extracting trips can reach 100 times.^[101]

Negel et al. designed a thin-disk multipass amplifier, which contains a mirror array of 40 mirrors for a total of 40 reflections, and a retroreflecting mirror pair to minimize the deflection. Combined with ZPL (969 nm) pumping, pulses of 1105 W, 7.3 ps, peak power of 0.18 GW,^[54] and 1420 W, sub-8 ps, peak power of 0.56 GW^[62] were obtained. Furthermore, by increasing the number of reflections to 60, a laser output of 2 kW and 6.7 mJ and peak power of 0.97 GW^[102] was achieved. Dietz et al. reported a multi-pass cell (MPC) with excellent mechanical and thermal stability for industrial use.^[93] The MPC includes 34 mirrors, and the beam spot in the MPC is unchanged. With pumping at 941 nm, the system produced an output of 600 kHz, 950 fs, 2.3 mJ, and a peak power of 2.28 GW. Based on a dual-wavelength pump of 940 nm and 961 nm and a two-stage MPC structure, the pulse energy was 4.8 mJ, and the peak power was 3.47 GW.^[53] This integrated MPC+CPA-free structure has great potential in commercial disk lasers.

2.2. From 10 GW to 100 GW

As mentioned above, pulse energy larger than 10 J and peak power of 1 GW were realized using active mirror slab amplifiers.^[48,83,89,90] With this distributed structure, Ogino et al. demonstrated a record power of 310 W using a conductive-cooled active-mirror amplifier with a pulse energy of 10 J. The master amplifier consisted of four large-aperture Yb:YAG ceramics (a diameter of 60 mm and thickness of 7 mm), liquid nitrogen cooled at 78 K, and PDL pumped.^[49] They aimed to scale the energy to 100 J with 10 GW peak power by combining more than 10 large-aperture laser heads. However, there is still no public report of more than 10 GW peak power by the active mirror.

Another excellent heat dissipation structure is a multislab geometry that divides the laser gain volume into multiple slices and extracts heat with the coolant flowing through the surface at high speed to force convection to transfer the heat from the surface to the coolant, which was first demonstrated in the Mercury system.^[103] Helium gas is commonly used as a coolant because it is chemically inert and has the lowest thermo-optic coefficient, which means it has a minor influence on beam quality. The medium gain aperture can be greater than 100 mm, and the thickness can be less than 10 mm.^[104] With multislab amplifiers, the world's first one kW-level average power^[50] and highest pulse energy of 250 J^[105] among LD-pumped nanosecond solid-state lasers were achieved.^[103] In addition to the larger crystal aperture, multislab also has the following advantages. 1) It can approach the optimum laser/pump light mode-matching efficiency. 2) The pump surface with the highest heat generation density is directly cooled, showing efficient heat dissipation capacity. 3) Thermal gradient along the optical axis is slight; it is helpful to realize high beam quality. The Mercury laser system built by LLNL in the United States obtained a pulse output with a single pulse energy of 61 J, a repetition frequency of 10 Hz, a pulse width of 15 ns.^[106] Its multislab configuration main amplifier stage includes two gain modules (amplifier heads). Each gain module contains seven parallel Yb:S-FAP slabs (40 mm $\times$ 60 mm $\times$ 7.5 mm) cooled by high-speed helium gas at room temperature.^[103,107] The pump laser subsystem of the High

Repetition Rate Advanced Petawatt Laser System (HAPLS) was developed by LLNL.^[108]

The helium gas cooling technology was adopted in a multislabs configuration as the Mercury system. The gain medium was Nd:APG-1, which yielded 70 J and 3.3 Hz output at 1053 nm.^[109,110] In 2017, the output was improved to 97 J.^[66] Further using cryogenic helium gas cooling technology, the DiPOLE laser system developed in the UK used 150 K helium to dissipate heat to the main amplifier. The laser had 105 J, 10 ns, 10 Hz, and 10.5 GW peak power with an average power of 1.05 kW.^[50] The laser facility provided pulse energy of 145 J and peak power of 14.5 GW in 2021.^[51] The abovementioned laser systems were developed as pump laser sources for PW class laser systems. For industry applications, Hamamatsu Photonics K.K. in Japan developed an industrial-grade nanosecond pulsed solid-state laser with a single pulse energy of 250 J and a pulse width of 30 ns that operates at a repetition frequency of 0.2 Hz. Each amplifying module adopted a multislabs structure with 10 Yb:YAG ceramics pieces. The temperature adjustment range of cooling helium gas was 150–300 K.^[105]

Since the pulse energy of the directly amplified multi-pass disk is at the mJ level, it is far from reaching the saturation fluence of thin-disk crystals.^[111] Therefore, it is feasible to adopt the low repetition frequency and high energy regenerative amplification technology.^[112,113] Compared with a multipass amplifier, the number of passes of reamplification can reach hundreds of passes, which can achieve high magnification up to 10^8 times.^[87,91] However, high peak power can easily damage intracavity devices. CPA solves the problem of obtaining ultra-intense pulses for application without any nonlinear processes destroying the amplifier. Combined with CPA, a peak power of tens of GW^[114] to ≈ 100 GW^[115] has been obtained with pulse energies up to 100 mJ by a thin-disk regenerative amplifier. Meanwhile, the beam quality is only related to the regenerative amplifier cavity.^[114,116–118]

However, when the pulse repetition period is on the order of the gain relaxation time, the regenerative amplifier may appear to have period-doubling bifurcations,^[119–125] which aroused widespread interest among researchers. Grishin and Michailovas systematically summarize continuously pumped solid-state regenerative amplifiers' dynamic characteristics and influencing factors. It has been proposed that increasing the seed pulse energy can improve the amplification dynamics, and adding a preamplifier is an effective suppression scheme.^[126]

Klingebiel et al. demonstrated a thin-disk regenerative amplifier with CPA and obtained 220 mJ of pulse energy at 1 kHz under PLD pumping of 940 nm.^[112] The average output power is scaled to 540 W at 10 kHz with 969 nm ZPL pumping, the maximum pulse energy was 50 mJ, and the pulse duration was <1.2 ps.^[127] Using a dual thin-disk laser-active medium, the system provided more than 100 mJ of pulse energy at a 10 kHz repetition rate.^[128] By increasing the repetition rate of the pulses to 20 kHz, the pulse energy was 95 mJ with a pulse duration of <600 fs.^[129]

Even if combined with CPA technology, Pockel cell damage remains an issue under high energy conditions. The output power of the thin-disk regenerative amplifier and the pulse's energy is limited by the aperture of the electro-optical crystal in the cavity, and the electro-optical device will introduce additional losses

(mainly depolarization). The maximum average power and pulse energy are limited by the damage threshold, available crystal size, and feasible switching voltage of the cell.^[98,116,130] To further increase the average power and pulse energy, a hybrid amplifier should be used. The thin-disk regenerative amplifier can be used for pre-amplification, and thin-disk multipass amplification is used for the main amplifier. In this way, the peak power of pulses can exceed 500 GW.^[81]

TRUMPF Scientific Lasers GmbH reported a hybrid amplifier system at 1030 nm for Laser Lightning Rod project. The seed pulses were stretched using a Chirped Fiber Bragg Grating (CFBG) and amplified to 240 mJ in the regenerative amplifier at 1 kHz, then boosted to a final energy of 800 mJ, a pulse width of 1 ns by four-stage thin-disk multipass master amplification. Finally, a compressed pulse of 920 fs with 720 mJ energy has been tested for long-range filaments generation.^[81]

However, the thin-disk amplifier is limited by its gain volume, and its output pulse energy is under the order of J.^[131,132] In order to achieve greater energy amplification, thicker disk or even mm-level thick AM amplifiers have been reported. Jung et al. obtained two synchronized chirped ns pulse with 80 mJ energy from a regenerative am. They seeded the pulses in a unique large aperture ring amplifier equipped with two 750 μm thick Yb:YAG (7%) disks of 25 mm diameter. Over 1.2 J pulse energy after 4 round trips and a compressed duration of 2.5 ps have been achieved.^[131]

Moreover, Wang et al. demonstrated a disk-slab composite structure based on cryogenically cooled master amplifiers which include two Yb:YAG active mirrors (the master mirror is a 30 mm $\times$ 30 mm square, 2 mm thick, 3%-at Yb:YAG), the output laser pulses of 1.1 J, 1 kHz with a pulse duration of 4.5 ps was achieved which was signaled by a 500 μm thick disk regenerative amplifier. This is the highest average power picosecond laser with an energy of joule-level reported to date. The wider 4.5 ps pulse width is caused by discarding part of the spectral width of the RA seed in order to overcome the temperature mismatch between the room temperature seed and cryogenic master amplifiers.^[64]

2.3. From TW to 10 PW

All record-holding ultrafast laser systems mentioned above are based on the relatively narrowband gain material such as Yb:YAG, Nd:YAG, which presents a hurdle in attaining sub-100 fs pulse duration.^[53,81] By using a wider gain bandwidth crystal such as Yb:CaF₂, Ti:sapphire, combined with CPA technology, or adopting a nonlinear parametric crystal (BBO, LBO, DKDP) for nonlinear frequency conversion, one can get several cycles of compressed pulse with the peak power of TW and even PW.

Compared with CPA-Ti:sapphire, the crystal of optical parametric amplification (OPA) does not store energy which avoids the gain narrowing effect and the parasitic lasing. In addition, the exceptional broadband of OPA provides enough bandwidth for ultrashort pulses. Since the transmission window of the conversion crystal can reach more than 2000 nm, a sub-three cycle fs pulse can be achieved.^[84,133,134] And even a sub-two cycle pulse width has been reported by a wider complementary spectral range.^[135] Currently, the CEP-stable (carrier-envelope phase), multi-TW, few-cycle OPCPA systems have been widely applied

in the relativistic light-matter interactions. Specifically, a contrast ratio of more than ten orders of magnitude is necessary to prevent the detrimental plasma formation;^[136] the pulse-width less than three cycles is required for the excitation of few-fs to attosecond XUV pulses and a stable CEP maintenance.^[137] In order to achieve the above-mentioned excellent performance, a high-energy disk laser with less than 1 ps pulse width and an mm-level short crystal is adopted straightforward.^[134] Besides, the pulse front matching between pump and signal is critical to high conversion efficiency.^[133]

Furthermore, when the peak value is scaled to PW-level, a large aperture of 100 mm is essential for the damage prevention of nonlinear crystals. Today, the most advanced crystal growth technology provides pore sizes for two types of nonlinear crystals: LBO and DKDP. Compared with LBO, the lower second-order nonlinearity and damage threshold limit the peak power at the sub-PW level.^[138] And over 1 PW peak-power output near 800 nm has already been reported by a hybrid CPA-OPCPA configuration which included a master OPCA stage with a large size LBO (100 mm × 100 mm × 17 mm).^[139] Moreover, a 3-cascaded all-OPCPA laser facility has generated a compressed pulse of 18.6 fs with 4.9 PW at 0.2 Hz.^[56]

Although there will be obstacles from gain narrowing and the parasitic lasing on the road to achieve a 10 PW breakthrough. Via the temporal dual-pulse pumped scheme and the index-matching liquid cladding technique, the parasitic lasing and the transverse ASE can be eliminated effectively. Besides, a spectral preshaping method can be applied to precompensate the spectral shifting and suppress the spectral gain narrowing. A recompressed process has also been adopted for optimizing pulse compression. In this way, the Shanghai Superintense Ultrafast Laser Facility successfully yielded a compressed pulse of 21 fs with a peak power of 10.3 PW pumped by a homemade Ti:sapphire laser in which the Ti:sapphire crystal has a diameter of 235 mm and thickness of 72 mm,^[140] and the latest output report is 12.9 PW.^[55] There are other PW facilities worldwide, and many laboratories have put forward plans to develop 10 PW laser systems and even EW-class laser systems.^[141] Some new technologies are expected to increase the peak power further. The compression after compressor method (CafCA) is based on spectral broadening by self-phase modulation (SPM) in nonlinear plates and eliminating the spectral phase by chirped mirrors to generate high peak power by further narrowing the pulse width. Some experiments were carried out where a 100 mm diameter 5.5 J pulse was compressed from 57 to 22 fs.^[142] Furthermore, a pulse with an energy of 18 J was compressed from 60–70 to 11 fs.^[143] Also, a pulse with an energy of 18 J was compressed from approximately 60 to 10 fs; the peak power reached 1.5 PW.^[144] Lasers with tremendous pulse energy and peak power can be achieved through time-domain pulse combined.

In addition to the recompression scheme, prior to the master-amplifier two subsequent temporal coherent combination strategies can be adopted to obtain a wider stretched pulse width. As the maximum pulse width stretched from the traditional CPA is about 10 ns limited by the available grating sizes. Promising techniques following this concept are divided pulse amplification (DPA) and coherent pulse stacking amplification. DPA is a time-domain pulse combining method. Linear delay lines are

employed in this DPA approach to make successive replicas of the stretched pulse, which are then recombined back to a single pulse using an identical delay-line configuration after amplification. Limited by the required number and length of the delay lines, the number of pulse replicas is usually less than 10 which makes it difficult for the final peak to reach TW.^[145,146] Hence, another alternative delay line can be implemented by a passive stack-and-dump cavity which is the core components of the pulse stacking amplification.^[147] Although coherent pulse stacking amplification is currently mainly used for pulse energy enhancement of high repetition frequency fiber lasers, it still has considerable reference significance for high repetition frequency disk lasers.

2.4. Enabling Technology

2.4.1. Gas and Cryogenic Cooling

Cryogenic cooling technology can be used to change the performance of the active medium to achieve high conversion efficiency. For the active medium of Yb:YAG, as the temperature decreases from room temperature to 80 K, the laser material changes from a quasi-three-level system into a four-level system, reducing the reabsorption of the laser.^[65] When cooled to a low temperature, the thermal conductivity of undoped YAG increased by nearly nine times relative to that at room temperature. Although this advantage is weakened when a moderate level of doping is introduced, there is still a significant improvement of three to five times.^[148] Both the thermal expansion coefficient and the thermo-optical coefficient are improved. When the temperature drops from 300 to 77 K, the former is reduced by four times, and the thermo-optical coefficient is reduced by seven times. In addition, the emission spectrum of Yb:YAG is significantly narrowed at 77 K, which leads to an increase in the stimulated emission cross-section and a corresponding decrease in the saturation fluence below 2 J cm^{-2} .^[111,149,150] Based on the cryogenically cooled Yb:YAG active mirrors and pulsed diode laser pumping, Wang et al. reported a laser pulse output of 1 kHz, 4.5 ps, 1.1 J, which is the highest average power Joule-level ps laser reported at that time.^[64] Based on four liquid nitrogen circulating cooled laser heads,^[49] Ogino et al. achieved a laser output of 9.3 J, 10 ns, at 33.3 Hz.

2.4.2. ZPL and PDL Pumping

The laser amplifier can use ZPL pumping or PDL pumping to directly reduce the heat input, thereby reducing the heat load of the active medium.^[151,152] Pumping directly to the laser upper energy level to reduce the thermal load of the active medium. For example, compared Yb:YAG with the 940 nm pump, the ZPL pumping in 969 nm can reduce heat generation by 32%. The pump power intensity can be increased by approximately 48% under the same thermal load conditions.^[63] In terms of thin-disk amplifiers, based on ZPL pumping, Negel et al. at the University of Stuttgart and TRUMPF Laser GmbH+Co. KG achieved a fundamental mode laser pulse of 800 kHz, 7.3 ps, and average power of 1105 W with an optical efficiency of 44%.^[54]

In the low-repetition, high-energy laser pulse amplifier, PDL pumping can significantly reduce the heat load on the active medium and suppress ASE, which leads to high conversion efficiency.^[58,61,89,90] The multi-slab master amplifier stage of HAPLS was jointly developed by LLNL and Lasertel; the pump system consisted of 4 LD arrays at 888 nm with peak power of 3.2 MW and pulse width 300 μ s,^[153] and a 97 J, 3.3 Hz laser pulse at 1053 nm has been achieved.^[66] As for thin-disk regeneration amplification, Jung et al. used a pulsed diode laser pump at 940 nm (1.7 J and 970 μ s) and a 100 Hz, 1.8 ps, 240 mJ pulsed light produced.^[113] It is worth noting that this incredible pulse energy is directly accomplished through a single disk regeneration cavity. The damage threshold of BBO crystals will limit the further increase of the peak value.

2.4.3. ASE Suppression

The energy storage capability of the active medium is limited by unavoidable spontaneous emission (SE). When the SE is amplified in the pumped medium, ASE is generated, one of the main limitations to obtaining high-energy pulses.^[87,154–158] When the pulse repetition period is on the order of the gain relaxation time, ASE is the main limiting effect on the energy rise.^[67] The main effects of ASE on laser amplification include depleting the population density of the upper laser manifold,^[68,154,156,159] generating parasitic lasing,^[160,161] significantly reducing the gain of the laser,^[87] and thermal distortion effects.^[154,162]

The Monte Carlo type ray-trace technique^[68,87,92,154,163,164] and the finite element analysis technique^[165,166] are commonly used to analyze the ASE effect and suppress the ASE.^[68,164,167–170] For the suppression of ASE, the primary purpose is to limit the oscillation of SE. The main methods currently used are anti-ASE cap,^[155,156,169,171–173] absorber cladding,^[90,174–184] beveling angle,^[67,69,163] and nonlinear gradient doping.^[185–188] The above-mentioned DiPOLE laser system applied an absorber cladding. The Yb:YAG gain slab was encircled by a 5 mm wide Cr-doped YAG to minimize ASE and suppress parasitic oscillations.^[104]

2.4.4. Stable Light Field for Disk

For thin-disk lasers, multipass magnification requires a stable ratio between the pump spot and the beam spot to ensure extraction efficiency and suppress the effects of self-focusing and thermal lenses.^[2]

Conventional 4f-imaging thin-disk multipass amplification is commonly performed in the early stage.^[189] The advantage of this structure is fewer devices for multi-pass amplification since the beam size can be reproduced regardless of the thermal lens. However, unlike the beam size, the phase front curvature and divergence accumulate linearly with the dioptric power, limiting the number of the amplification-trip and the output beam size.^[67] Sometimes a vacuum environment was needed to prevent optical breakdown due to the air ionization.^[190]

Another Fourier-based structure can turn the phase distortion into a spatial halo, and part of the halo is lost through the soft-aperture effect, which provides excellent beam quality.^[191] A hybrid architecture combined with the 4f-imaging and

Fourier-based principle can achieve high gain in a compact arrangement.^[192] Based on this structure, the spherical phase front distortions occurring at the active medium can be compensated, and a small signal gain of 30 has been achieved through 20-pass exaction. Moreover, the total propagation distance is significantly shortened with slightly increased system complexity.^[193]

Utilizing a concave radius of disk and the plane mirror array, the beam can stay almost collimated with large beam size through the propagation. This way, tight intermediate foci were effectively eliminated and contributed to smaller accumulated beam deterioration and B-integral.^[54] Due to the large fundamental mode matching on the disk crystal over a short space distance, it is currently the commonly used structure for the high-average-powered ultrafast lasers.^[52,53,93] However, the system is more sensitive to the variation of the focal length of the thermal lens; therefore, the thin disk needs to have a weak focus function to prevent the light beam from gradually spreading out in free space transmission.^[96]

2.4.5. Correction of Wavefront Distortion

Unlike disk amplifiers, wavefront distortion always appears for high thermal stresses generated by pumping conditions and transverse temperature gradients in the gain medium of the high-energy slab system.^[73,194,195]

The wavefront distortion is automatically corrected via phase conjugation, eliminating the requirement for active electrical component control. High-energy solid-state lasers have been successfully used with liquid Freon SBS PCMs.^[74] The SBS in Freon has a higher threshold and requires a longer working distance to achieve good reflectivity in the acoustic grating form. As a result, it can only work with a pulsed laser with a greater coherence length. In most cases, the PCM's reflectivity is less than 1, necessitating using a high-gain structure to avoid a reduction in efficiency. Adaptive optics (AO) provides a wavefront correction control method that is more flexible and easier to implement than phase conjugation.^[75] The complexity of active control loops and actuators increases due to this capacity. To decrease or eliminate the optical path difference of the high-power output beam, the adaptive optics system actively or adaptively controls the form or direction of the optical components in the beam path. Even if thermal gradients were reduced to a small fraction of the overall thermal change in optical path length, this would typically still be enough to cause wavefront distortion to degrade the beam quality. The wavefront distortion of the HALNA laser system is corrected by the SBS phase conjugate mirror (length 800 mm, FC-77 fluorinated liquid), which can achieve 60% of the output energy concentrated within two times the diffraction limit of the far-field.^[194] The wavefront distortion of a 5 J, 6.6 ns, 200 Hz solid-state Nd:YAG laser was corrected using an adaptive optics system, according to Yu et al.^[73] The PV value (peak-to-valley) of the wavefront is lowered from 26.47 to 1.91 μ m after automated correction, and the beam quality is enhanced from 18.42 to 2.86 times the diffraction limit. The PV value of the wavefront is less than 0.45 μ m after additional correction with a deformable mirror, and the beam quality is 1.64 times that of the diffraction limit.^[73] For the wavefront correction of the PW laser, two stages of deformable mirrors were employed. After wavefront

correction, the Strehl ratio was approximately 0.4, and the spot size was close to the diffraction-limited value.^[195]

2.5. Overview of Solid System

The diode laser pumped solid-state laser systems are summarized in **Table 1**. **Figure 2** show the pulse energy versus repetition rate and peak power versus repetition rate of various techniques. It is seen that the active mirror slab amplifier reached pulse energy of 10 J and a peak power larger than 1 GW.^[48,83,89,90] With multiple active mirror slab amplifiers, 100 J energy of ns pulse laser is under construction.^[49] Further, single pulse energy larger than 100 J, a pulse width of roughly 10 ns, a repetition frequency of 1–10 Hz, and a peak output of 10 GW can be produced from multislab amplifiers with PDL pumping and gas cooling technology.^[50,51,104,105,200]

As depicted in **Figure 2b**, the disk amplifier is the current standard method for producing high-energy ps laser pulses. Because of its outstanding heat dissipation performance, it is feasible to generate high-power ultrashort pulses. The average power of the thin-disk amplified ps pulses is beyond 2 kW.^[53] A laser-based thin-disk amplifier has been claimed to have 720 mJ and 0.92 ps at a 1 kHz repetition rate.^[81] A laser system based on cryogenically-cooled and PDL-pumped Yb:YAG generated 1.1 J pulses of 4.5 ps at a repetition rate of 1 kHz, with an average power of 1.1 kW, setting a record for Joule-level ps lasers.^[64] The few-cycle fs laser, based on a thin-disk high energy ps laser with OPCPA, has a peak output of more than 5 TW.^[133,134] Despite its energy storage limitations, disk lasers may provide multi-joule outputs at several kHz by multiplexing the number of gain amplification elements to distribute the heat load across numerous disks and continuing advancements in different components for extremely high-power levels (optics, dispersive elements, etc.)^[205]

As shown in **Figure 2b**, the fs laser-based CPA-Ti:sapphire, has a peak power of more than 10 PW, and numerous 10 PW lasers are already in development.^[141] OPCPA offers the advantages of having a large gain bandwidth, high gain, small thermal effect, and increased pulse contrast when reducing pulse length for intensity scaling. The fs laser, which is based on a high energy ns laser with OPCPA, has surpassed 5 PW,^[56] and building a 100 PW OPCPA laser facility has recently begun in Shanghai.^[206,207] An EW class laser was conceptually designed.^[208] When larger crystals, higher damage threshold gratings, and other optics become accessible shortly, a peak power greater than the EW laser might be created for the fundamental study. It is anticipated that employing the compression after compressor technique and other technologies to reduce the pulse width further would result in higher peak power.

Although a single fiber channel cannot achieve high peak pulse output due to insufficient energy storage and strong nonlinearity, this shortcoming will be overcome by the coherent beam combining. The most ambitious system proposed by ICAN (International Coherent Amplification Network) for average power would do coherent beam-combining of >10 000 fiber lasers to produce >0.1 PW peak power.^[209]

Next section, we will focus on how to achieve high-power amplification in the single-channel fiber amplifier and review the work done by researchers on the path of nonlinear suppression.

3. High Power Fiber Amplifiers

Optical fiber has a great specific surface area, ensuring the compact hundreds of thousands of watts fiber-optic system.^[210] This tightly bound fiber waveguide provides high-quality beam quality output and significantly enhances the effective nonlinear action length L_{eff} of the laser. Although this powerful nonlinear effect has opened up many new areas which are difficult to reach by solid-state lasers, including Raman amplifier with extended wavelength or self-cleanup;^[211–213] low noise amplifiers by stimulated Brillouin scattering (SBS) and four-wave mixing (FWM);^[214,215] supercontinuum excitation from vacuum ultraviolet to mid-infrared;^[216–220] and extraordinary dynamic phenomena of spatiotemporal soliton;^[221–223] etc. The nonlinearities often need to be suppressed effectively at most high-power CW and pulsed lasers resonators and amplifiers. Moreover, the self-focusing damage of the material seems to be the ultimate obstacle to limiting the further increase of peak power. However, the self-focusing threshold of quartz is around 4 MW for the fiber with a diameter of 100 μm . To eliminate the self-focusing damage, the peak power of the stretched ns pulse is limited to the sub-MW level, and the peak power of the compressed fs pulse has exceeded 5 GW. On the other hand, the failure of fiber lasers often occurs due to the thermal damage of the coating layer of the fiber and by the optical damage at the front-end device due to the returning light of the amplifier. Therefore, the key to overcoming these material damage challenges lies in suppressing nonlinear deterioration.

Among these nonlinear effects, SBS will only affect narrow linewidth ns pulses. Self-phase modulation (SPM) will stretch the pulse during the ultrashort pulse amplification process. Still, it can be significantly suppressed by CPA with phase-compensation and will not limit the peak power of the final compressed pulse. However, SRS will reduce the conversion efficiency of the continuous or pulse amplification process and induce mode instability.^[224,225] As the Primary nonlinear effect, SRS suppression is imminent.

3.1. SRS Suppression

The conventional definition of the SRS threshold was obtained by Smith.^[226] Subsequent researchers revised the threshold formula under the premise of considering many practical factors.^[227–229] Besides, Wang,^[230] Schreiber et al.,^[231] Liu et al.^[232] systematically analyzed the Raman evolution of high-power double-cladding fiber (DCF) oscillators and amplifiers, then gave some proposals of laser configuration: Reducing the effective length of the active fiber by manufacturing larger-mode field and higher-doped gain fiber, redshifting the signal wavelength, flattening the pump distribution by tandem pumping or side-pumping,^[233] increasing the FBG bandwidth in the kW-level continuous optical resonator appropriately, and reducing the signal power in the master oscillator power amplifier (MOPA) system etc.

3.1.1. Spectrally Selective Fibers

In addition to the above conventional adjustments, people have also developed some targeted design Raman filter fibers for the

Table 1. High energy and high peak power solid-state laser systems.

Structure	Peak power	Energy	Duration	Repetition rate	Year ^[refs.]
Sub - GW					
Rod-type	0.69 GW	9 J	13 ns	10 Hz	2017 ^[86]
Zigzag slab	0.76 GW	5 J	6.6 ns	200 Hz	2017 ^[72]
Active mirror slab	1.74 GW	13.9 J	8 ns	2 Hz	2013 ^[89]
	0.85 GW	8.5 J	10 ns	1 Hz	2014 ^[196]
	0.1 GW	1 J	10 ns	100 Hz	2015 ^[197]
	1.22 GW	12.2 J	10 ns	10 Hz	2017 ^[90]
	1.28 GW	12 J	9.4 ns	10 Hz	2019 ^[83]
	1.03 GW	10.3 J	10 ns	10 Hz	2019 ^[48]
	0.93 GW	9.3 J	10 ns	33.3 Hz	2021 ^[49]
Thin-disk multipass	0.18 GW ^{b)}	1.38 mJ	7.3 ps ^{a)}	800 kHz	2013 ^[54]
	0.56 GW ^{b)}	4.7 mJ	8 ps ^{a)}	300 kHz	2015 ^[62]
	0.97 GW ^{b)}	6.7 mJ	6.5 ps ^{a)}	300 kHz	2016 ^[102]
	2.11 GW ^{c)}	2 mJ	0.83 ps ^{a)}	200 kHz	2017 ^[52]
	0.699 GW ^{b)}	5.8 mJ	7.8 ps ^{a)}	300 kHz	2021 ^[198]
	2.28 GW	2.3 mJ	0.95 ps	600 kHz	2018 ^[93]
	3.25 GW ^{b)}	4.8 mJ	1.3 ps ^{a)}	400 kHz	2020 ^[53]
10 GW–100 GW					
Multislab	4.07 GW	61 J	15 ns	10 Hz	2008 ^[106]
	4.85 GW	97 J	20 ns	3.3 Hz	2016 ^[66]
	1.08 GW	10.8 J	10 ns	10 Hz	2015 ^[199]
	10.7 GW	107 J	10 ns	1 Hz	2016 ^[104]
	10.5 GW	105 J	10 ns	10 Hz	2017 ^[50]
	14.5 GW	145 J	10 ns	10 Hz	2021 ^[51]
	2.93 GW	117 J	40 ns	0.05 Hz	2020 ^[200]
	8.33 GW	250 J	30 ns	0.2 Hz	2021 ^[105]
Thin-disk regenerative amplifier	7.3 GW	2 mJ	0.21 ps	100 kHz	2016 ^[116]
	108.8 GW ^{b)}	220 mJ	1.9 ps ^{a)}	1 kHz	2015 ^[112]
	41.23 GW ^{b)}	50 mJ	1.14 ps ^{a)}	10 kHz	2016 ^[127]
	106.60 GW ^{b)}	110 mJ	0.97 ps ^{a)}	10 kHz	2017 ^[128]
	127.5 GW ^{b)}	240 mJ	1.77 ps ^{a)}	0.1 kHz	2016 ^[113]
	18.8 GW ^{b)}	20 mJ	1 ps ^{a)}	5 kHz	2016 ^[114]
	179.3 GW ^{b)}	206 mJ	1.08 ps	5 kHz	2017 ^[115]
	689 GW ^{c)}	720 mJ	0.92 ps	1 kHz	2020 ^[81]
	332.65 GW ^{c)}	1100 mJ	2.91 ps	1 kHz	2020 ^[64]
TW - PW					
OPCPA	5.5 TW	53.8 mJ	8.8 fs	1 kHz	2017 ^[84]
	4.9 PW	91.1 J	18.6 fs	0.2 Hz	2017 ^[56]
	5 TW	42 mJ	6.9 fs	10 Hz	2018 ^[133]
	4.4 TW	42 mJ	8.3 fs	100 Hz	2020 ^[134]
	4.8 TW	32 mJ	6.6 fs	1 kHz	2020 ^[85]
	0.6 PW	11.8 J	19 fs	/	2021 ^[138]
CPA/Ti:sapphire	0.5 PW	16 J	28.6 fs	3.3 Hz	2017 ^[201]
	4.2 PW	83 J	19.4 fs	0.1 Hz	2017 ^[202]
	0.3 PW	10 J	30 fs	0.1 Hz	2018 ^[203]
	10.3 PW	217 J	21 fs	1 Hz	2018 ^[140]
	12.9 PW	300.9 J	23.4 fs	one shot per 3 min	2020 ^[55]
CPA/ Yb:CaF ₂	0.17 PW	16.7 J	98 fs	0.02 Hz	2016 ^[204]

^{a)} (Retrieved pulse duration obtained from the autocorrelation trace.); ^{b)} (Peak powers have been calculated assuming a Gaussian pulse shape (Peak power = 0.94 energy/duration);^[99]) ^{c)} (Peak powers have been calculated assuming a sech²-pulse shape (Peak power = 0.88 energy/duration).^[81])

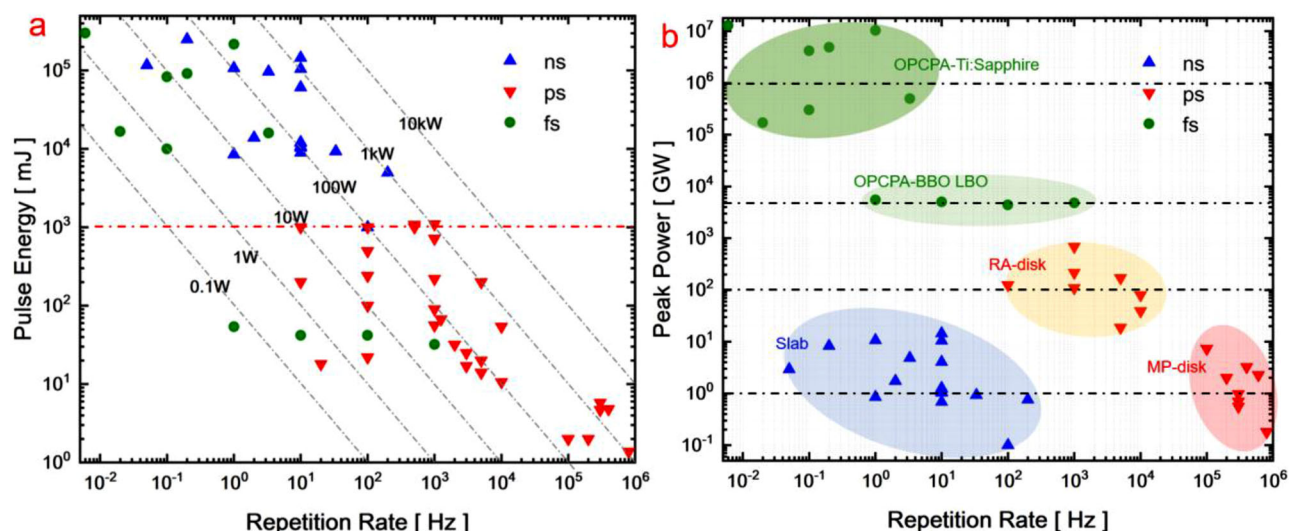


Figure 2. Summaries of high-energy ns/ps/fs lasers. a) Pulse energy versus repetition rate. b) Peak power versus repetition rate. RA (regenerative amplifier), MP (multi-pass).

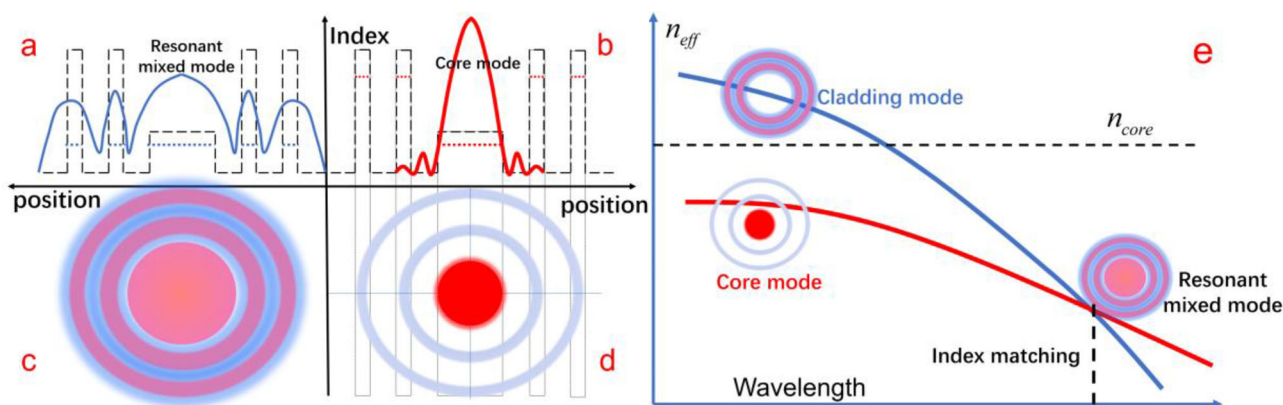


Figure 3. Illustrate the filtering concept due to the resonant coupling: The black dashed line in (a,b) represents the refractive index distribution. a) When the effective refractive index (blue dotted line) is equal, the resonant coupling process occurs otherwise. b) The core mode (red line) is maintained. c) Mixed-mode distribution. d) Core mode distribution. e) The effective refractive index for the two modes. a, b, e) Reproduced with permission.^[236] Copyright 2005, Optical Society of America.

characteristics of significant Stokes shift (≈ 13 THz) of silica. Kim et al. designed a W-type refractive index profile fiber.^[234] The cut-off wavelength of the optical fiber was set between the signal and the Stokes wavelength. Under moderate bending, the intense bending loss of the Stokes wave can be produced while maintaining a weak bending loss of the signal. Besides the bending-induced loss filtering mechanism,^[235] a resonance coupling process can occur by introducing cladding sidebands with a higher refractive index than the core region.^[236,237] When the Raman effective refractive index at the core and the ring region is equal in Figure 1a. Then the Raman mode presents a high-loss resonance mixed-mode distribution in Figure 3e.

However, delocalization cannot eliminate the mixed mode. Only by completely vetoing the existence of the Raman mode from the waveguide restriction conditions can the Raman mode be suppressed more effectively without signal loss. Due to flexible bandgap design, the photonic bandgap fiber (PBGF) can act as

a sharp-cut distributed filtering for 1st Raman,^[238] ASE,^[239] and high-order Stokes.^[240] A detailed description of photonic crystal fiber (PCF) and PBGF will be in chapter 3.5.

3.1.2. LPGs and CTFBGs

The filtering mentioned above fiber is limited in its high-power application due to its bending sensitivity and insufficient adaptability for diameter scaling. But this design concept can be directly implemented on the gain fiber and inspired the subsequent fibers design, e.g., updating the high-refraction sideband to the high-refraction side core.^[241]

Jansen et al. simulated that inserting several lumped filters can achieve a threefold increase in the Raman threshold with reasonable insertion loss.^[242] Nodop et al. first proved this concept through CO₂ laser-induced long-period fiber grating (LPG),

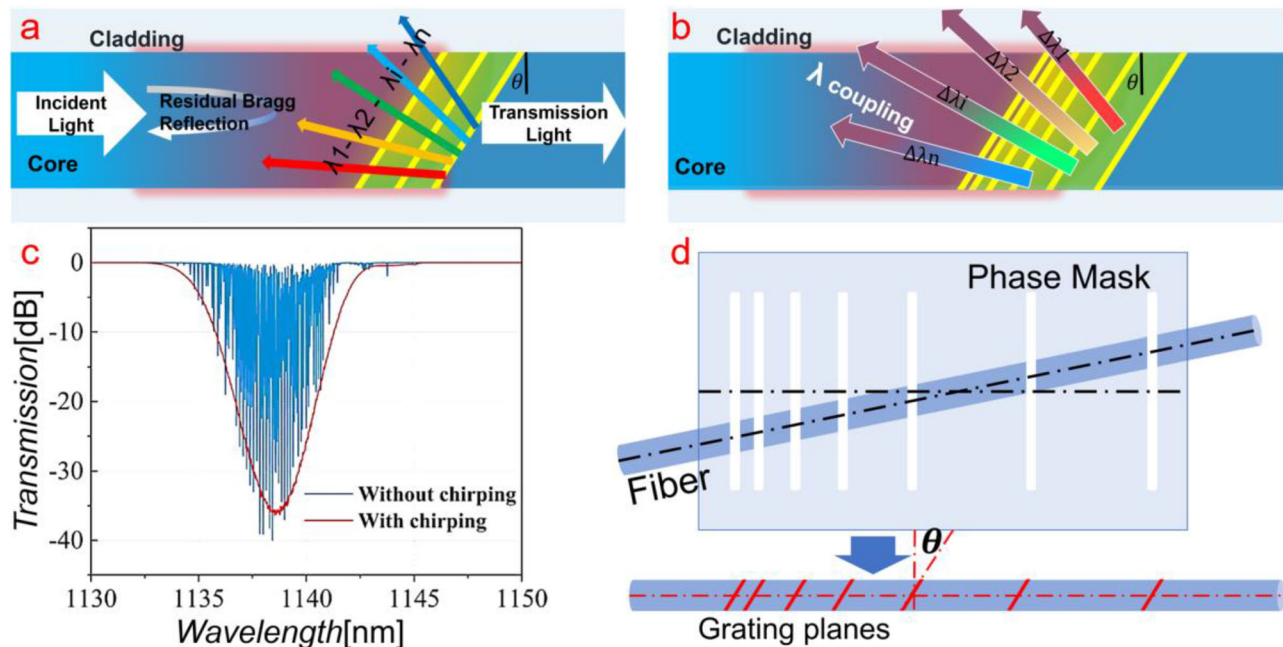


Figure 4. Schematic diagram of the CTFBG. a) Titled FBG without chirp: solid colored arrows represent the separated resonance wavelength λ_i . b) Titled and chirped FBG: gradient filled arrows illustrate the coupled spectral broadening under chirp modulation. c) Simulated transmission spectra of a titled FBG and a CTFBG. Reproduced with permission.^[253] Copyright 2019, Optical Society of America. d) Schematic of rotating phase mask method.

three LPGs cascaded to double the extractable output power.^[243] However, the etching only occurs on one side of the fiber surface due to the high absorption of carbon dioxide by quartz. The asymmetry and random thermal stress result in temperature, strain, and bending sensitivity, limiting the high-power working capability.^[244–247] Although fs direct transmission etching of large-core fiber is performed, the filter spectrum results still need to be optimized.^[248,249] Combined with the dynamic annealing process; Insensitive packaging, and Gaussian apodization, the LPGs etching of 14/250 fiber has been realized, and it has been directly applied to the kW-level MOPA system to achieve a 24 dB suppression ratio with side-lobes eliminated.^[250] Unlike typical point-to-point manufacturing technology, Hu et al. also reported a more stable printing etching technology by a galvanometer. The working power is still at a low level of 100 W, limited by the fiber diameter.^[251]

Another type of grating with a more stable filtering performance is the chirped and tilted fiber Bragg gratings (CTFBG), which Meltz first proposed in 1990.^[252]

Thanks to the tilted grating plane (parallel yellow slash segments in Figure 4a,b), the incident light will couple not only to the backward core mode just like conventional FBG, which is called the residual Bragg reflection, but also to various orders of cladding modes with different coupling angles and resonance wavelength λ_i . As each cladding mode has a narrow bandwidth and is separated only considering the effect of grating tilt, the titled FBG without chirp presents a perfect “comb”-like transmission spectrum, as shown in Figure 4a. Suppose the grating is chirped in Figure 4b. In that case, the periods of the titled FBG at different positions will have the effect of broadening each cladding mode resonance to such an extent that they end up overlapping considerably (red line in Figure 4c).^[253] CTFBG is fabri-

cated by rotating the phase mask, as shown in Figure 4d. The chirped phase mask and the optical fiber (previously processed by high-pressure hydrogen for weeks) are fixed at a relative angle θ . The main spectral information can be tailored by adjusting the original Bragg wavelength, tilt angle, chirping rate, and grating writing time.^[254]

Liu et al. introduced chirp into a titled FBG and realized the design and inscription of a CTFBG with broadband filtering for the first time.^[254] Wang et al. fabricated a CTFBG (under Corning HI1060 sm fiber) to suppress the SRS in a 1090 nm fiber laser. A maximum suppression ratio of nearly 25 dB is demonstrated after inserting the filter between the seed and SM amplifier with a total pump power of 36 W.^[255] Furthermore, they successfully realized that inscribing the filters on the LMA-GDF-10/130-M and PS-GDF-20-400-M fiber, with >10 nm bandwidth and ≈ 20 dB suppression ratio, have been accomplished through higher hydrogen loading pressure and longer time.^[256] They inserted the CTFBG filters between the seed and the kW-amplifiers and compared the Raman suppression in copumping or bi-pumping;^[257] and tandem pumping situations.^[258]

The fiber gratings made of photosensitive fibers have relatively worse power tolerance than the traditional optical fiber, which limited the filter can only be inserted between the low power seed and master amplifier like CTFBG1 in Figure 5. Two main obstacles must be overcome when CTFBG2 is serviced at the final power output position. First, the dynamic annealing process can eliminate the strong infrared absorption caused by the residual hydrogen molecular after the high-pressure hydrogen treatment. Second, profound heat accumulation in the coating induced by the coupled Stokes light.

Jiao et al. adopted a segmented-corrosion cladding power stripping technology to mitigate the overheating. In this way, the

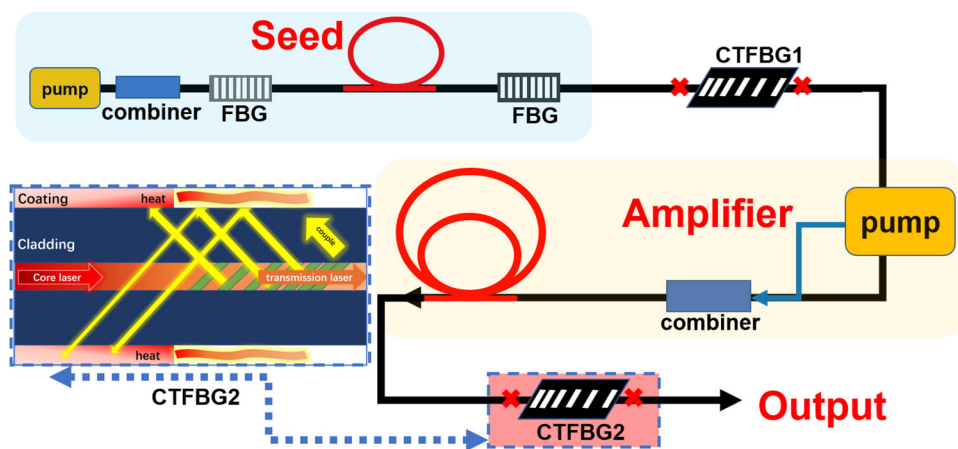


Figure 5. Schematic diagram of interstage filter CTFBG1 and direct filter CTFBG2, Inset shows the heat accumulation under the coating.

direct suppression of SRS for kW power was achieved first. The thermal slope is lower than $0.015\text{ }^{\circ}\text{C W}^{-1}$, with a 23 dB SRS suppression ratio.^[253] Aimed at several kilowatt-level direct filtering, Song et al. manufactured multiplexed CTFBGs composed of five continuous segments of FBGs with weaker exposure duration and larger power capacity. The independent gratings are etched on the LMA 25-400 fiber and then spliced one by one to form a composite filter grating. By integrating them into a MOPA system of a new recorded 3.4 kW output, extra SRS suppression of $\approx 15\text{ dB}$ has been obtained at 1134.5 nm, and the temperature rise of the multiplexed CTFBGs device registered at the maximum output was only about $5\text{ }^{\circ}\text{C}$.^[80]

3.2. SBS Suppression

The narrow linewidth fiber laser with near diffraction limit beam quality is desired in various applications, such as non-linear frequency generation,^[259,260] remote sensing,^[261,262] beam combination,^[263–266] gravitational wave detection.^[267] SBS effect is the leading factor of limiting power scaling in narrow linewidth continuous wave or nanosecond fiber laser amplifiers. Once the peak power reaches the threshold of SBS, the energy of the signal begins being transferred to the frequency-downshifted (about 11 GHz in silica) Stokes wave with spreading on the contrary direction to the signal, which destroys the stability of the amplifier or oscillator, and even causes physical damage to the devices.^[76,268] In recent years, many SBS suppression schemes have been verified theoretically and experimentally by taking advantage of the narrow gain bandwidth of SBS ($\approx 10\text{ MHz}$) and the characteristics of its threshold. We will expand the description according to the classification in **Figure 6**.

3.2.1. Changing Brillouin Frequency Shift

Since the Brillouin shift in the fiber is related to the acoustic velocity at the pump wavelength and the effective mode refractive index, adjusting these parameters can change the Brillouin Shift. If the frequency shift is greater than the SBS gain bandwidth, the Stokes wave will cease to grow because of the substantial decrease of the Brillouin gain.^[268] This method can be realized in

the following two ways. One is to employ fibers to reduce the overlap of the acousto-optic modes.^[269] Different doping materials will cause different acoustic and optical speed responses,^[270] the core and cladding of fiber can be doped in a partition to achieve the separation of acoustic mode and optical mode, enhancing the SBS threshold.^[271–274] The other one is to control the peak Brillouin gain by applying the influence of environmental factors (such as stress, temperature, etc.) on the Brillouin frequency shift.^[275,276] A 3.3 dB SBS suppression ratio has been reported by temperature gradient control at $70\text{ }^{\circ}\text{C}$.^[277]

3.2.2. Controlling the Interaction Length

The interaction between Stokes wave and signal in SBS is the key to the whole process, and SBS can be inhibited by effectively regulating the length of interaction. For a continuous laser, the interaction length is the length of the fiber, and SBS can be reduced by employing a short gain fiber with a large mode field diameter. Unlike continuous lasers, the interaction length in pulsed lasers is no longer simply equal to the fiber length but related to fiber length, pulse width, pulse repetition frequency, etc. Su et al. used the MOPA system combined with the 0.31 m short interaction length in the main amplifier stage to suppress SBS and produced a pulse width of 3 ns, an average power of 913 W.^[278] Huang et al. studied the linearly polarized all-fiber MOPA system with narrow-linewidth. Since the phonon lifetime of SBS is limited to 10 ns and the short fiber can reduce the interaction length between Stokes wave and signal, the average power of 466 W, a pulse width of 4 ns, the repetition rate of 10 MHz, and linewidth of 203.6 MHz was achieved.^[279]

3.2.3. Phase Modulation Technique

Although the SBS effect can be effectively suppressed by changing Brillouin frequency shift and regulating the interaction length between Stokes and signal, the impact of these two methods is limited in the power scaling of a multi-kilowatts laser system with narrow linewidth. In recent years, phase modulation technology has been an attractive solution to suppress SBS. By

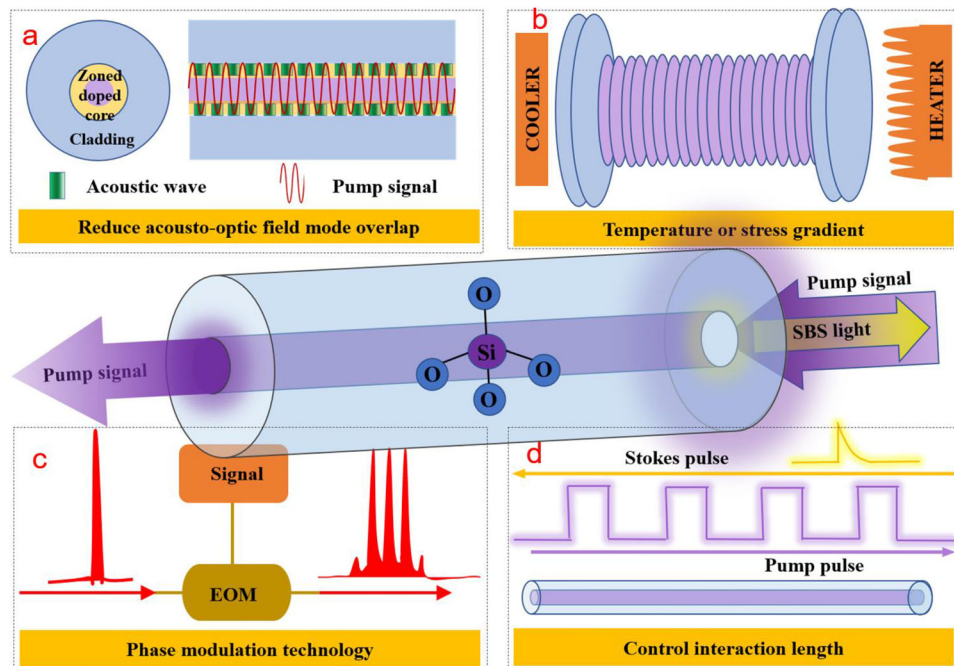


Figure 6. SBS effect and main inhibition schemes. a) Reduce the mode overlap of the acousto-optic field. b) Apply temperature and stress gradient to change the Brillouin frequency shift (the fiber is wound from one end of the rod to the other, and the temperature gradient along the fiber is created by using heaters and coolers at the end of the rod). c) Broaden the linewidth of laser seed by phase modulation. d) Control the interaction length between Stokes wave and pump signal.

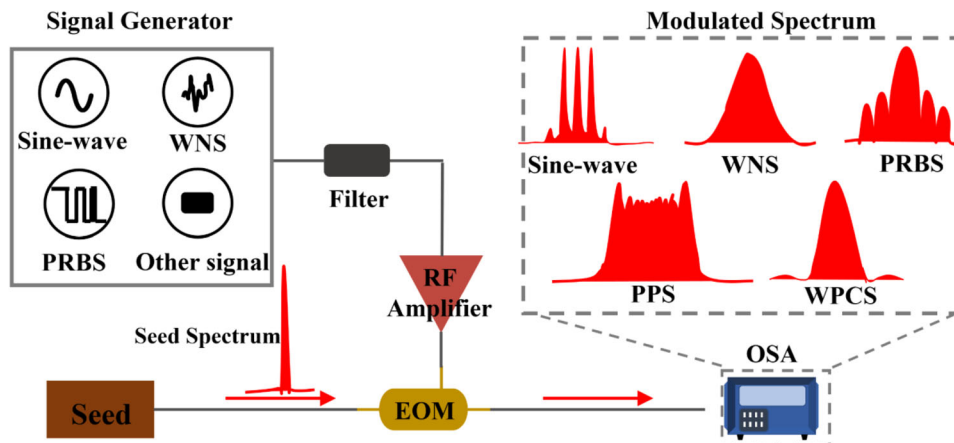


Figure 7. Schematic diagram of laser phase modulation technology. After filtered by filter and amplified by radio frequency (RF) amplifier, the signal drives the electro-optic modulator (EOM), which increases the line width of the seed laser. The sine-wave modulation produces equal-amplitude side frequency. The envelope after WNS modulation is Lorentz line type. The envelope after PRBS modulation is sinc^2 type, which contains separated spectral lines. For the PPS modulation, the spectrum is approximately rectangular. The spectrum after multi-phase coded signal modulation is similar to the sinc^2 type, but the density is higher than that of PRBS modulation.

applying a signal to the seed phase to broaden the linewidth, the power spectral density of the seed laser can be reduced, which can obtain effective suppression of SBS in the frequency domain.^[280] With the vigorous exploration of researchers, the modulation signals currently used for phase modulation are pretty abundant, mainly including sine-wave phase modulation, white noise signal (WNS) modulation, pseudo-random bit sequence (PRBS) modulation, piecewise parabolic signal (PPS) modulation, multi-phase

coded signal modulation, etc. Among these signals, WNS and PRBS are the most used in multi-kilowatt-level fiber amplifiers with narrow-linewidth.^[281–283] **Figure 7** shows the processes of the phase modulation technique.

Sine wave, WNS, and PRBS modulation are classic methods to suppress SBS. Sine-wave modulation generates side frequency by applying a sinusoidal signal to the phase of the seed laser through a signal generator. The modulation spectrum has two main

characteristics: it shows equal-amplitude spectral lines with multifrequency. Both the spectral width and the number of spectral lines are proportional to the modulation amplitude.^[284] Limited by the broadening ability of a single modulator, multiple sine-wave modulation signals are often used to increase the number of spectral lines and achieve spectral broadening.^[285–287] Williamson et al. proposed a scheme to suppress SBS by WNS phase modulation. The filtered WNS signal is amplified by a radio frequency amplifier. It drives the electro-optic modulator, which can carry out different phase shifts of seed laser to broaden the spectral width.^[288,289] Li et al. applied cascaded WNS phase modulation to widen the linewidth of seed laser to tens of GHz, and Raman signal-to-noise ratio reached 49 dB at the output power of 1275 W.^[290] Through comparing laser amplification systems with and without WNS phase modulation, Lin et al. found that cascade WNS phase modulation technology is beneficial to inhibit SBS and SRS with the output of 0.3 nm line width and 3.7 kW power.^[291]

Sine wave and WNS phase modulation can broaden the linewidth of seed laser >20 GHz to restrain SBS. Still, they also cause problems such as degradation of beam quality due to dispersion during spectral beam combining.^[292] Therefore, SBS suppression needs to be performed under a narrower linewidth. PRBS is another phase modulation signal which has been widely concerned in recent years. The modulation depth of 0 or π is applied to the seed laser by generating a series of “0” or “1” levels using a shift register with length N . Since the continuous number of “0” and “1” generated by the system is pseudorandom, the phase shift applied to the seed laser is not the same at different time.^[293] The modulated spectrum of PRBS shows periodic discrete optical frequency lines in the sinc² envelope, and the optimization of spectral line spacing can further improve the SBS threshold.^[294] By employing a line spacing of 12.7 MHz and linewidth of 2.2 GHz, Yang et al. obtained a backward reflectivity of <1‰ at an output power of 1.2 kW. The results showed SBS threshold increased with the decrease of spectral line spacing but reached saturation when spectral line spacing was 10 MHz.^[292] In addition, they investigated PRBS phase modulation techniques in the form of maximal-length sequence (MLS). They found SBS inhibition is nonmonotonically reliant on mode length, modulation depth, etc., to maximize SBS suppression.^[295]

In recent years, researchers have been pursuing a better inhibition effect of SBS and balancing some problems in this process, such as SBS TMI in high power expansion, coherence weakening caused by linewidth broadening of the seed laser. Therefore, some new phase modulation and optimization schemes have been developed.^[296,297] Lai et al. proposed multi-phase coded signal phase modulation, which changed SBS thresholds superior to WNS and PRBS modulation in fiber amplifiers with long active fibers pumped at 915 and 1018 nm. They believed that multi-phase coded signal modulation would play a key role in balancing SBS and TMI.^[298] White et al. applied a PPS phase waveform to generate a higher SBS threshold per unit bandwidth. If the bandwidth is defined as a frequency range containing 85% of the total power, the SBS threshold of PPS phase modulation is 1.4 times higher than that of the 5- or 7-bit PRBS modulation scheme. Besides, the SBS threshold for PPS is three times higher than PRBS, with bandwidth containing 95% of the total power.^[299] Al-

Table 2. Representative research results of fiber laser with narrow linewidth.

Linewidth	Year ^[ref]	Power [kW]	Linewidth [GHz]	Pulse width [ns]
<1 GHz	2016 ^[305]	8.5	0.22	3
	2018 ^[279]	8.8	0.203	4
	2019 ^[306]	4.02	0.129	8
	2019 ^[303]	75	0.33	2.8
	2021 ^[307]	30	0.283	3.8
	2021 ^[304]	31.4	0.063	8
1–10 GHz	2015 ^[286]	3.9	4.5	3.5
	2019 ^[308]	0.818	<7	CW
	2020 ^[292]	1.27	2.2	CW
>10 GHz	2016 ^[309]	2.9	82	CW
	2017 ^[310]	2.43	68	CW
	2018 ^[290]	3.5	48	CW
	2019 ^[311]	2.19	23	CW
	2019 ^[312]	1.5	13	CW
	2019 ^[291]	3.7	79.5	CW
	2021 ^[281]	4.92	154.3	CW
	2021 ^[282]	3.5	47.8	CW
	2021 ^[283]	3.25	20	CW

though phase modulation has made remarkable achievements in suppressing SBS, a broader linewidth is needed with a higher power or longer transmission fiber, ultimately resulting in invariable or reduced laser power spectral density and weakening laser coherence.^[300,301] Goodno and Rothenberg broadened the linewidth of seed laser by importing time-independent phase with WNS and synchronous amplitude and phase modulation with a sinusoidal signal. The nonlinear self-phase modulation in the power scaling is applied to compress the spectrum, and the SBS-limited linewidth is reduced by 2.5 times.^[302]

Based on these suppression schemes, the narrow linewidth fiber laser has made remarkable achievements (shown in **Table 2**, where the power means the peak power for pulsed laser). The peak power of multikilowatt has been realized under the linewidth of GHz, for example, 1.27 kW was generated with a linewidth of 2.2 GHz, a 3.7 kW with a linewidth of 79.5 GHz have been reported.^[291,292] More recently, laser output with a linewidth of 154.3 GHz and power of 4.92 kW has also been achieved.^[281] At the MHz level, such as a linewidth of 330 MHz, peak power of 75 kW, pulse duration of 2.8 ns, and another example of the linewidth of 63 MHz, peak power of 31.4 kW, and a pulse duration of 8 ns is also realized.^[303,304]

3.3. Transverse Mode Instability

As mentioned above, a larger core diameter is a straightforward way to suppress most nonlinear effects. Meanwhile, it is necessary to reduce the numerical aperture (NA) of the LMA fiber to achieve a single transverse mode output. This means that the refractive index values of the core and the cladding need to be as close as possible. However, the high doping of the core naturally opposes the slight refractive index gradient and homogenous refractive index profile. When the fiber is highly doped by Yb, other

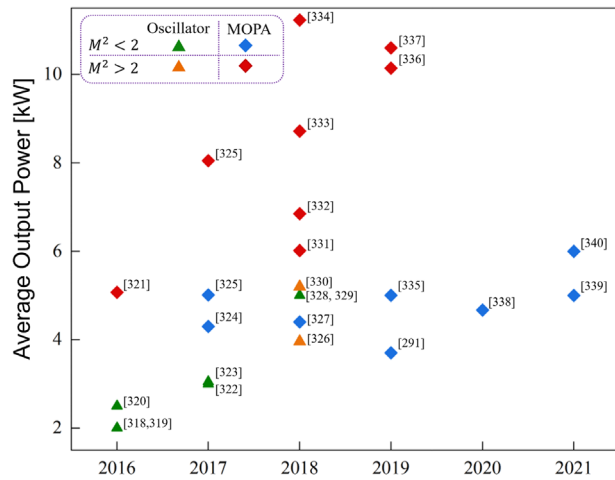


Figure 8. The development of fiber laser average power for Oscillator and MOPA systems.^[291,318–340]

elements, such as Al and Ce, must be doped to suppress the photodarkening effect. These comprehensive considerations bring many challenges in the preparation of LMA fibers with low NA, such as high optical quality of the fiber core, high doping uniformity, high precision refractive index modulation, and low light attenuation as possible. The minimum NA parameter achieved by the traditional MCVD process is 0.05, so the maximum core diameter of traditional step index fiber (SIF) is 16 μm .^[313] Benefitting from the development of the manufacturing process, the current minimum NA value can reach 0.02. LMA with mode field diameter (MFD) of 35 μm and a NA of 0.025 (even 0.02 NA) could be made, the system gives an output of 1.6 kW and the M^2 value is 1.04.^[314] A $\text{Yb}^{3+}/\text{Al}^{3+}/\text{F}^-$ codoping LMA with 50 μm MFD and 0.02 NA was made, and the output M^2 value was <1.1 by coiling fibers with a diameter of 0.35 m.^[315]

However, even 0.02 NA does not support a pure SM output with a core size exceeding about 40 μm , so LMA fibers for suppressing nonlinearity usually support multiple transverse modes. This, combined with the thermal load of the fiber, leads to the phenomenon of TMI. TMI appears that when the average output power reaches a certain threshold (usually between 100 W^[316] and thousands of Watts^[317]), the output beam quality and beam stability will suddenly deteriorate. Since TMI may have occurred before the output power reaches the nonlinear threshold, it has become a significant factor limiting the power expansion of SM and quasi SM fiber lasers. The outputs of single module fiber lasers in the recent five years are shown in **Figure 8**. The power of SM and near SM output fiber laser increases slowly as the development goes on compared with multi-modes output fiber laser due to the TMI effect.

3.3.1. Mechanism and Threshold of TMI

The earliest description of the TMI phenomenon comes from the PCF amplification system.^[317] Then in 2011, Eidam et al. reported visualized laser outputs with unstable modal contents that fluctuated over time and deterioration of beam quality at threshold output power. This phenomenon is named “Transverse

Mode Instability.”^[316] The scientific research community generally agrees that it originates from the thermal effect, and it is a manifestation of stimulated thermal Rayleigh scattering in fibers.^[341–345]

There are two conditions for the occurrence of TMI. One is the thermal-induced long period refractive index gating (RIG) produced by the modal interference pattern (MIP) introduced by fundamental mode (FM) and high-order mode (HOM) with different phase velocities. The other is the phase shift between MIP and RIG because the change of MIP is instantaneous, but the change of RIG falls behind MIP due to the time delay caused by heat conduction in the generation of RIG. By satisfying two conditions for TMI generation, energy coupling between the FM and the HOMS can be realized.

For optical fiber systems with large mode fields and short-length fibers, the process of TMI is composed of the stabilization stage, transition stage, and chaos stage, which can be seen in **Figure 9**. During the transition stage, the fluctuation of the beam profile occurs periodically on the millisecond time scale. This fluctuation can be identified as discrete frequency peaks in the spectrum shown in Figure 9a. Utilizing photodiode (PtD) measurements, the relevant time and dynamic information can be easily obtained and analyzed by Fourier transform. This leads to a widely used definition of the TMI threshold, which is reached when the local derivative of the fitted evolution curve (black line in Figure 9c) of the standard deviation of the PtD trace with the average power becomes $0.1\% \text{ W}^{-1}$.^[343] In addition, there are also other threshold definition standards like a constant average heat load.^[346]

However, there is no definite conclusion about the initial phase shift between MIP and RIG. There are two popular theories. The first holds a center frequency difference $\Delta\nu$ between the FM and the HOMS that generates MIP, and the frequency detuning causes the MIP to move axially in the optical fiber.^[341,347] Another theory is that there is no frequency difference between the FM and the HOMS. When the average power is high enough to make the heat load reach a certain level, the heat load will induce the change of waveguide conditions resulting in a difference of effective refractive index between the FM and HOMS, and the above process leads to the movement of MIP.^[348,349]

The two theories all have developed respective numerical simulation methods. These numerical models have been expanded to include factors such as transverse spatial hole burning (TSHB), gain saturation, quantum noise, pumping and signal modulation, signal bandwidth, and so on, which can influence the TMI threshold.^[347,350–353] They can also reproduce the precise physical details of various TMI processes through visual images.^[342,345,348] These factors point out the ways of inhibiting TMI.

3.3.2. Mitigation Strategies of TMI

In general, the parameters that affect the threshold of TMI can be divided into thermal effects and nonthermal factors, and the most important are thermal effects. The thermal load in optical fibers mainly comes from quantum defect (QD) and photon darkening (PD). According to the generation conditions of the TMI, reducing the heat load is the primary starting point for suppressing it. Research shows that the thermal load is not only from QD but

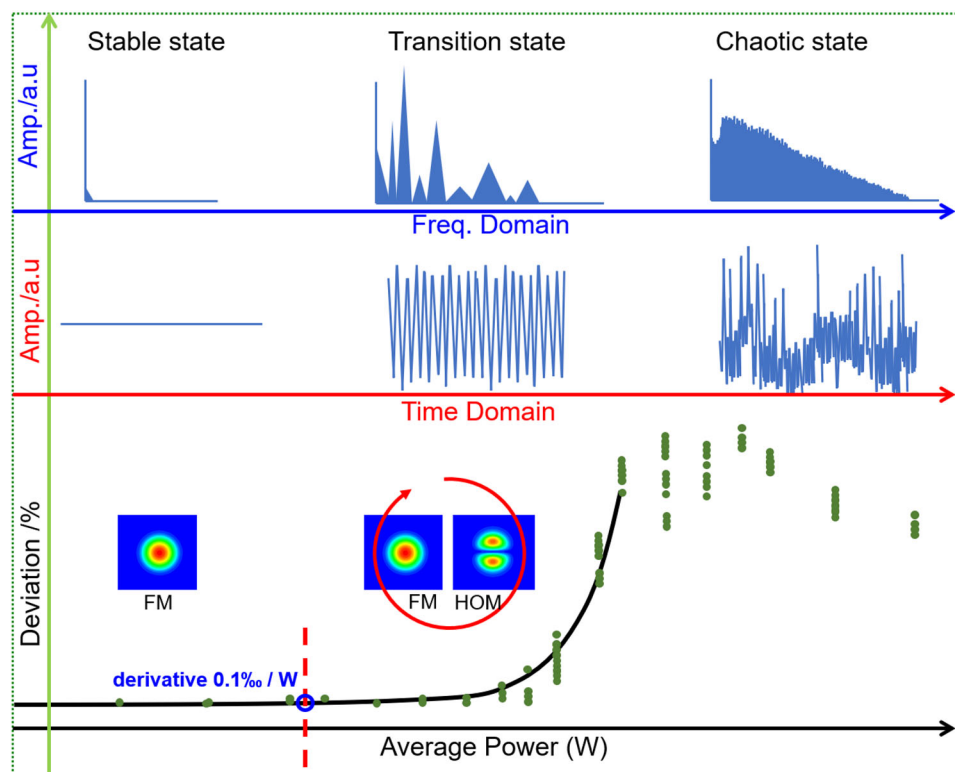


Figure 9. Steady-state stage corresponding trace of PtD shows a straight line and stable beam profile. After TMI Threshold, output comes into transition stage, TMI occurs, PtD trace shows strong and regular oscillations (the corresponding Fourier transform has specific peaks), the beam profile fluctuates periodically with time. The output shows a complex beam profile at the chaotic state, and PtD traces irregular oscillation with a broader peak in the frequency domain. Reproduced with permission.^[343] Copyright 2012, Optical Society of America.

the contribution of the PD heat load is more evident than that of imagination. QD transforms only a small amount of pump light into heat while PD transforms almost all the absorption of the pump and signal light into heat.^[346,354–356]

However, further studies have shown that the reduction of QD by a single pump using a higher absorption cross-section often has the opposite effect.^[357] This is because the high effective absorption pump injection will significantly reduce the effective amplification length of the amplifier, which is a powerful way to reduce the nonlinear accumulation of the system, but it strengthens the thermal gradient distribution of the fiber.^[356] Therefore, considering the total heat load, it is also necessary to reduce the thermal gradient.

Enhancing gain saturation reduces the thermal gradient laterally, and intense gain saturation induced TSHB will “erase” the thermally induced refractive index grating in the back half of the fiber, reducing the stimulated Rayleigh scattering and the energy exchange between the FM and the HOMs.^[342,350] Simultaneously, the intense consumption of inverted populations can also suppress the thermal effect brought by PD. Hansen et al. found that the TMI threshold obtained when the gain saturation effect is considered three times the calculation result, ignoring the numerical model’s gain saturation.^[352]

Using the gain saturation effect means that the effective length of the fiber will increase, which will lead to the emergence of various nonlinear effects, especially the SRS effect. Recent studies have found that the emergence of SRS will lead to the early oc-

currence of TMI. Hejaz et al. found that the generation of TMI sometimes closely depended on SRS.^[224] The increase of Raman output components above the SRS threshold output power would destroy the SM spot, and SRS inhibition can effectively inhibit TMI. In 2021, Distler et al. held that the thermal load of the Raman conversion process affected the TMI threshold.^[358] For the same fiber amplifier configuration, the TMI threshold appears at approximately the same Stokes output power, and the threshold value is independent of the Raman conversion efficiency.

In addition to factors related to thermal effects, the modulation (or noise) of the pump and signal, quantum noise, self-heating Rayleigh scattering, and signal light bandwidth can also affect the TMI threshold.^[347,351,353,359] Stihler et al. found that a pump noise component close to the TMI fluctuation frequency (usually <10 kHz) can reduce the TMI threshold about three times.^[359] Smith et al. separately assumed quantum noise, pump light modulation, and signal light modulation in their simulations. Letting these influencing factors bring the injection of HOMs seed components detuned to the fundamental mode frequency in the 1 kHz frequency band, HOMs seed can significantly reduce the TMI threshold.^[347] they also studied the effect of high-order mode seed components caused by self-heating Rayleigh scattering. The study found that the self-heating Rayleigh scattering seed has a more substantial influence than quantum noise, which may affect the TMI threshold of the amplifier.^[351]

Otto et al. used an acousto-optic polarizer to actively control the phase shift in the optical field in the fiber to destroy the

formation of RIG in fiber, thereby increasing the TMI threshold.^[360] Different from this spatial structure, Montoya et al. controlled the mode excitation through the all-fiber photon lantern, and the TMI threshold was increased from 800 W to 1.27 kW.^[361] Jauregui et al. found that a higher pump power corresponds to a smaller period of the MIP in the gain fiber and a shorter period of thermal-induced refractive index grating. When the pump light is periodically modulated with appropriate frequency and amplitude, the formation of the thermal index grating in the gain fiber can be destroyed (or called “cleaning”), which will weaken the mode coupling and increase the TMI threshold.^[362] In this way, the TMI threshold increased from 260 to 560 W. But the output power is also strongly modulated due to changes in pump power. Another active inhibition scheme to wait for exploration is to control the phase shift of MIP and RIG and achieve the “beam cleaning” brought by positive phase-shifting. The energy flows from HOMs to FM caused by positive phase shift is called beam cleaning, and it could happen at any average output power.^[349] The latest numerical simulation shows that the degraded beam could return to the near diffraction limit state by injecting a traveling wave into the amplifier to control the modal energy flow.^[363]

Although many active and passive suppression methods have been studied, the origin of TMI is ultimately rooted in the existence of high-order modes in optical fibers. The most common ways to control mode externally are fiber locally tapering and coiling.^[364,365] The coiled active fiber of YDF 20/400 under bidirectional pumped MOPA configuration has achieved a narrow linewidth SM 4.92 kW output at 1071 nm by 976 nm pumping,^[366] and the SM power scaled to 6 kW with a larger size fiber of YDF 25/400 by 981 nm pumping.^[340] However, it is easy to cause partial overheating by pouring a large amount of core laser into the cladding through the tapered end cone.^[364] Too small coiling will reduce the mode field area of the fundamental mode and introduce the loss of FM. It is challenging to suppress TMI and SRS at the same time without changing the basic properties of optical fiber.

Currently, considering the development limitation of large mode field low NA SM fibers, people have designed many fibers with variable structure gain. One way is to improve SIF fibers with restricted doping to prioritize the gain fundamental mode while ensuring that the fiber core has a flat refractive index at the same time. Using a single-trench fiber, one can simultaneously increase the mode field and achieve the loss of high-order modes. Multiple-trench fibers and various structural variants can be used to realize high-order mode delocalization and leakage channel effects.^[367–370] Other optical fibers, such as tapered double cladding fiber (TDCF) and chirally coupled core fiber (CCCF), also improve the TMI threshold by introducing HOM transmission loss. Some new structures have also been developed on photonic crystal fibers, such as large pitch photonic crystal fiber using thermally induced refractive index precompensation to achieve high-order mode delocalization and LP01 mode restrictions.^[371] Leakage channel optical fiber (LCF) realizes high-order mode leakage and high loss.^[372] To completely eliminate the influence of PD, Nd:YAG crystal can be used as the primary material for fiber.^[373] In addition to starting from various sources that affect TMI, we can also choose to bypass TMI, such as using multicore optical fiber.^[374]

However, the manufacturing process limits the mode field of confined doped fiber and the average power based on this scheme is about 1 kW.^[375] and the crystalline fiber, multi-core fibers are difficult to be promoted due to high processing costs. The following section will selectively introduce the functional fibers (CCCFs, TDCFs, PCFs) widely used in several kW-level average power or MW-level peak, high-power amplification systems.

3.4. CCCF and TDCF

3.4.1. Single Mode Maintenance of CCCFs

The CCCFs contain a central main straight fiber core with one^[241] or multithin spiral side cores.^[79,376,377] Different from the traditional weakly coupled parallel waveguide system, effective selective HOM coupling between the central and the side cores satisfies the quasi-phase matching condition which can be realized by specific helix pitch Λ and side core offset radius R as shown in **Figure 10a**.^[378] HOMs in the central core will be coupled into the side core and thus depleted, leaving only the fundamental mode LP01 for stable transmission in the large central body.

The excellent SM output characteristics of CCCFs were tested by the S^2 method.^[379] The experiment performed by NLIGHT company compared the fiber mode content between LMA 25/250 μm fiber and 55/400 μm Yb doped CCCFs.^[79] The measurement results of LMA 25/250 (Figure 10b) clearly show that there are several additional peaks in different group delay curves, which means that the output beam still includes HOM contents even with Gaussian distribution profile. However, even with a larger core diameter, 55/400 μm CCCF can maintain almost pure SM content in Figure 10c.

Therefore, CCCFs are very suitable for single mode maintenance amplification. Compared with traditional large-mode field fibers, it has a more stable long-term fundamental mode maintenance capability and higher TMI threshold.^[381,382] There are also some new side core design concepts to reduce the manufacturing difficulty of classic thin side core or achieve a stable orbital angular momentum for multiplexing communications.^[383,384]

3.4.2. Nonlinear Suppression of CCCF

Because of the particular chiral coupling structure of CCCFs, each mode in the central core has a different quasi-phase-matching with different wavelength positions and coupling strengths. By the design of parameters, the wavelength selective loss caused by the lossy resonance of the central core mode can introduce stokes wave suppression. Theoretical research and numerical simulation show that the dependence of the Raman threshold on fiber length is effectively eliminated. CCCFs can achieve distributed and $>10 \text{ dB m}^{-1}$ stokes wave loss.^[385] Hu et al. have experimentally tested SRS suppression by importing nanosecond pulses of 30 kW peak power into a 6 m long CCCF with 33 μm core size and over 50 dB signal-to-noise ratio have been achieved.^[386] NLIGHT also used a 55/400 μm CCC fiber to build a laser amplifier. The laser output with a peak power greater than 1 MW and a pulse width of 50 ps and $M^2 \approx 1.2$ is obtained. The SRS signal is not observed when the spectral signal-to-noise ratio exceeds 50 dB.^[79]

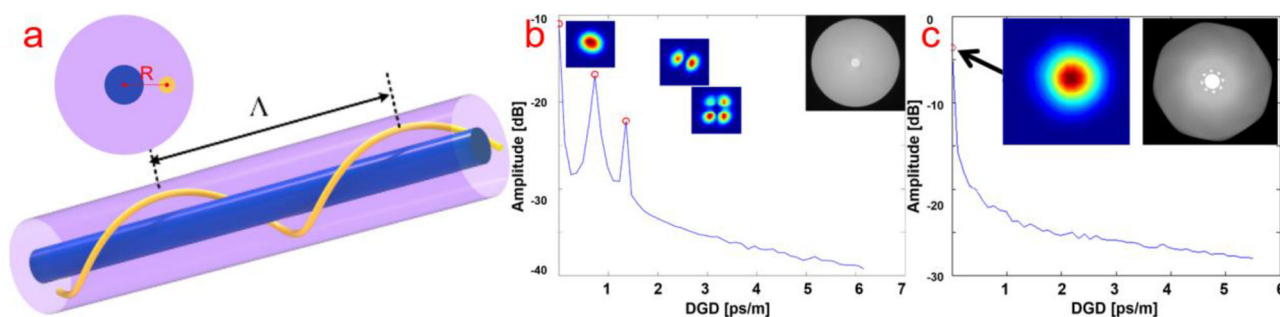


Figure 10. a) 3D structure and the cross section and 3D structure of CCCF. Reproduced with permission.^[380] Copyright 2018, Optical Society of America. b) Cross section and S^2 result of traditional LMA 25/250 μm fiber. c) 55/400 μm CCCF. Reproduced with permission.^[79] Copyright 2015, Society of Photo-Optical Instrumentation Engineers (SPIE).

The large permissible central core of the CCCF also benefits SBS suppression. Zhu et al. used a 2.7 m long 37/260/440 μm CCC fiber as the amplification stage to achieve laser output up to 511 W ($\Delta\nu < 40$ MHz), no SBS signal was observed.^[387] Hochheim et al. used a 34/250 μm CCCF to realize single frequency line biased TEM_{00} mode laser output with a center wavelength of 1064 nm and average output power of 100 W.^[388]

Besides, ultra-low relative power noise SM fiber laser for gravitational wave detectors using (GWD) CCCF amplification is being studied, and the relative power noise of the CCCF amplifier system could be up to two orders of magnitude lower.^[389,390] CCCF also shows great potential in the pulse coherent pulse stacking amplification and coherent beam combining. Pei et al. used an 85 μm core CCC-fiber based system to amplify 81 ns chirped burst pulses, then combined pulsed coherently and compressed into a single <540 fs pulse with multi-mJ.^[391] Recently, in 2021 Rainville et al. combined pulses coherently from three CCCFs into a single 22 mJ pulse. The combined pulse bursts match the proper shape for equal nonlinear phase across individual pulses and therefore are compatible with coherent pulse stacking and ultrashort pulse generation.^[392] This work represents an important step to expand to ≈ 100 mJ in the short term and to >1 J in the future.

3.4.3. kW-Level CW Laser of TDCF

Although the specially designed longitudinal uniform fibers can suppress TMI by introducing HOMs loss, the gain saturation and the low absorbable skew modes still remain residual at the fiber tail.^[393] On the one hand, counter-pumping can suppress gain saturation but requires a corresponding combiner and a supplementary reverse stripper.^[394] On the other hand, various cladding design schemes have been adopted to destroy the axial symmetry of the optical fiber cladding to eliminate the weakly absorbing pump modes.^[395–397] This does not eliminate saturation sufficiently and is only suitable for small-core fibers.^[398] In addition, a segmented amplifier has also been designed. However, the local step-taper welding will introduce significant insertion loss and beam quality deterioration by nonadiabatic mode conversion.^[399–401]

People invented a long-tapered fiber based on the above experience that satisfies the adiabatic tapering condition.^[402–404] This

longitudinally nonuniform tapered waveguide can not only realize stable amplification of the FM of the signal but also significantly reduce the beam quality requirements of the reverse injection pump (>300 mm mrad).^[405,406] The nonreciprocity produced by this geometric asymmetric waveguide exhibits natural ASE and SBS suppression.^[407,408] Overall, the tapered DCF can directly strengthen mW-level signal with weak nonlinearity and good beam quality, just like an amplifier chain group with continuously changing core diameter.^[409]

Tapered fiber opens a new way for high-power fiber amplifiers. At present, it has been adopted in kW-level all-fiber resonators,^[410] MOPA amplifiers,^[411] MW-level CPA-free amplification,^[409,412] and high-power single-frequency (SF) laser output.^[413,414]

Filippov et al. manufactured and systematically studied the critical parameters of tapered fiber both theoretically and experimentally.^[406,415] By optimizing the taper ratio, longitudinal profile, pump absorption coefficient, and core/cladding diameter ratio, the vignetting loss has been reduced to 0.45%. Then the power of 750 W with near theoretical limit slope efficiency of 81.9% has been achieved. A modest tapering rate and a slightly parabolic shape with high paraxial absorption are proposed.^[416]

For traditional long-tapered fibers, efficient pump conversion often requires spatial counter-pumping by a dichroic mirror. However, Shi et al. constructed an all-fiberized MOPA system based on copumping ytterbium-doped TDCF.^[417] When a 100 W seed power and 1970 W pump were launched, the output power at 1080 nm reached 1.47 kW with a slope efficiency of $\approx 72\%$. Even though the conversion efficiency is relatively low, this result shows the adaptability of the co-pumping scheme. Yang et al. also adopted the same co-pumping MOPA configuration based on a 20/400–30/600 μm linearly tapered fiber. The maximum output power was scaled to 1.72 kW at 1080 nm with an M^2 factor of ≈ 1.2 when pumped at 915 nm, with a higher TMI threshold than 976 nm.^[418] Furthermore, Ye et al. scaled the TMI-free power to 2.17 kW. They concluded that TDCF has a more effective TMI suppression ability than conventional uniform fiber with the same effective core diameters.^[419]

Besides classic monotonically tapered fibers, different longitudinal profiles in the fibers (as shown in Figure 11) have been manufactured for all-fiber kW-level MOPAs and oscillators.

Utilizing the ASE inhibition ability of TDCFs, Aleshkina et al. reported a monolithic oscillator by a novel saddle-shaped YDF. A

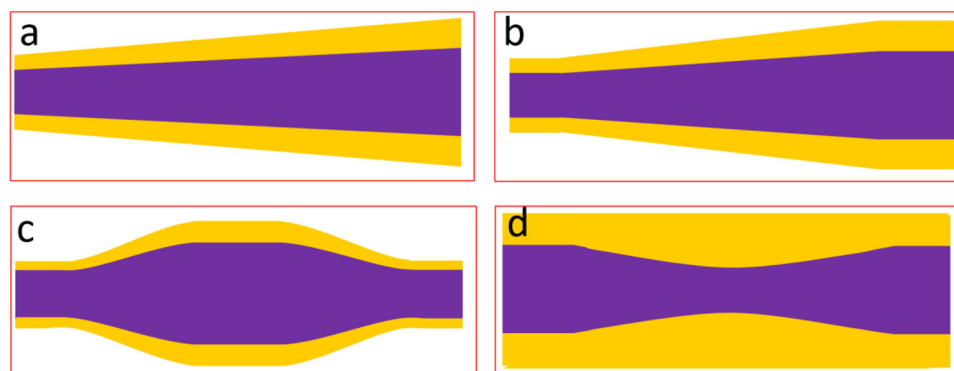


Figure 11. Schematic diagram of different types TDCF a) Linear tapered through the whole length. b) Linear tapered with a smooth port at both ends. c) Spindle-shaped core. d) Saddle-shaped core.

record SM output power of 10 W at 976 nm has been obtained by good suppression of the ASE at 1030 nm.^[420] Taking advantage of the SRS and TMI suppression ability, Zeng et al. achieved 1.3 kW output at 1080 nm by a saddle-shaped TDCF oscillator with 915 nm co-pumping.^[421] Moreover, 1.8 and 3 kW TMI-free output have been achieved using the improved bidirectional pumping under an all-fiber spindle-shaped oscillator.^[410,422] Finally, the power was scaled to 5 kW ($M^2 \approx 1.9$) by slightly increasing the fiber size and adopting a MOPA configuration. This is the highest CW power record by TDCF, the relatively low optical-to-optical conversion efficiency of 66.6% is expected to be further improved by optimizing the longitudinal fiber profile.^[411]

3.4.4. Nonlinear Suppression of TDCF

The amplification of short pulses is one of the most potential applications of TDCF. To achieve high peak power output, the tapered fibers must cooperate with the counter-pumping strategy, making the master gaining occur in the thick part to ensure an overall weak nonlinear amplification. In this way, one can build a CPA-free ps laser system capable of delivering MW-scale peak power,^[409] few tens of μJ pulse energy, and over 100 W power handling^[423] due to direct amplification of the low power signal. Bobkov et al. have demonstrated an 8 ps pulse amplification to a peak power of up to 0.76 MW, and a higher peak was limited by SRS.^[424] Furthermore, 150 W average power and 0.74 MW peak power for 22 ps pulses were generated through reasonable vignetting damage prevention.^[425] Petrov et al. reported a compact ps MOPA system that can yield 1.26 MW peak power and 200 W average power while preserving good beam quality.^[409] In addition to the long tapered type, a local adiabatic taper has also been utilized to achieve 2 MW peak power with a near-diffraction-limited beam quality of $M^2 = 1.3$, an inspiring record for ps amplifiers.^[412] The taper strategy also makes sub-MW peak power with mJ-level pulse energy possible for ns lasers. Roy et al. have obtained a 2.2 ns pulse with 1.1 mJ energy and 500 W average power, which confirms the power scaling ability of TDCF.^[423]

The continuous chain of amplifier model with a gradient core diameter is beneficial to high-efficiency amplification. It has a natural, effective SBS suppression ability for frequency shift in the SBS gain peak.^[408] The potential of the T-YDF for a high-

power SF fiber laser is experimentally demonstrated and theoretically analyzed. Patokoski et al. achieved a record-breaking 4 kW peak power and 524 μJ pulse energy of 130 ns; the output SF laser linewidth is <10 MHz with $M^2 = 1.08$.^[413] Huang et al. reported a 300 W stable linearly polarized SF output with a 9.3 MHz linewidth. Both ends of the PM T-YDF have a uniform core size, and the TMI effect in such an SF amplifier is thoroughly investigated.^[78] Furthermore, they demonstrated nearly transform-limited pulses with a linewidth of ≈ 283.8 MHz, pulse duration of 3.8 ns, maximal power of 8.8 W, and a record peak power of 30 kW, which symbolizes the excellent nonlinear performance of the TDCF.^[426] Lai et al. obtained a record average power of 550 W of an all-fiberized SF (linewidth of 45 MHz) fiber laser with a high slope efficiency of 80% and good SBS suppression. The power scaling was limited by the TMI.^[414]

3.5. PCF

PCF, as a kind of microstructure fiber usually made of a single material, is distinguished by the periodic air-hole array running along the whole length of the fiber. In contrast to conventional SIFs, the arresting feature of PCFs is the “endless single-mode” (ESM) behavior,^[427] i.e., remaining monomode at all wavelengths, which can be understood by the modal-sieve concept, as shown in Figure 12a,b.^[428] The fundamental mode cannot leak through the gaps between the air holes (diameter d , spacing Λ) due to its excessively large effective transverse wavelength component. The high-order modes are the opposite. Numerical results in Figure 12c reveal that proper manipulation of geometry, typically $d/\Lambda < 0.4$, promises that only the fundamental mode is supported.^[428,429] Extensive mode-area fibers with low nonlinearity and strict SM characteristics become possible, which can be employed to modify the performance of a high-power fiber system.

3.5.1. Large Mode Field

As fiber fabrication technology develops, such as stack-capillary-drawing,^[430–432] modified chemical vapor deposition,^[433,434] direct nanoparticle deposition,^[435] powder sintering,^[436] and

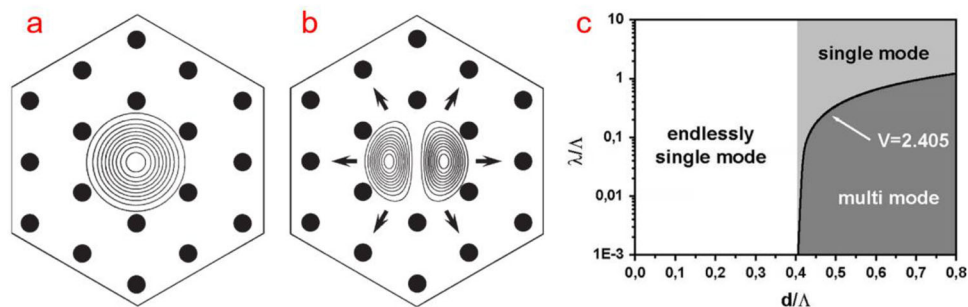


Figure 12. Schematic diagram of the sieve model. a) The fundamental mode is trapped. b) HOM leak away through the gaps between the air holes. Reproduced with permission.^[428] Copyright 2006, IEEE. c) Modal characteristics of a one-hole missing PCF. Reproduced with permission.^[429] Copyright 2006, IEEE.

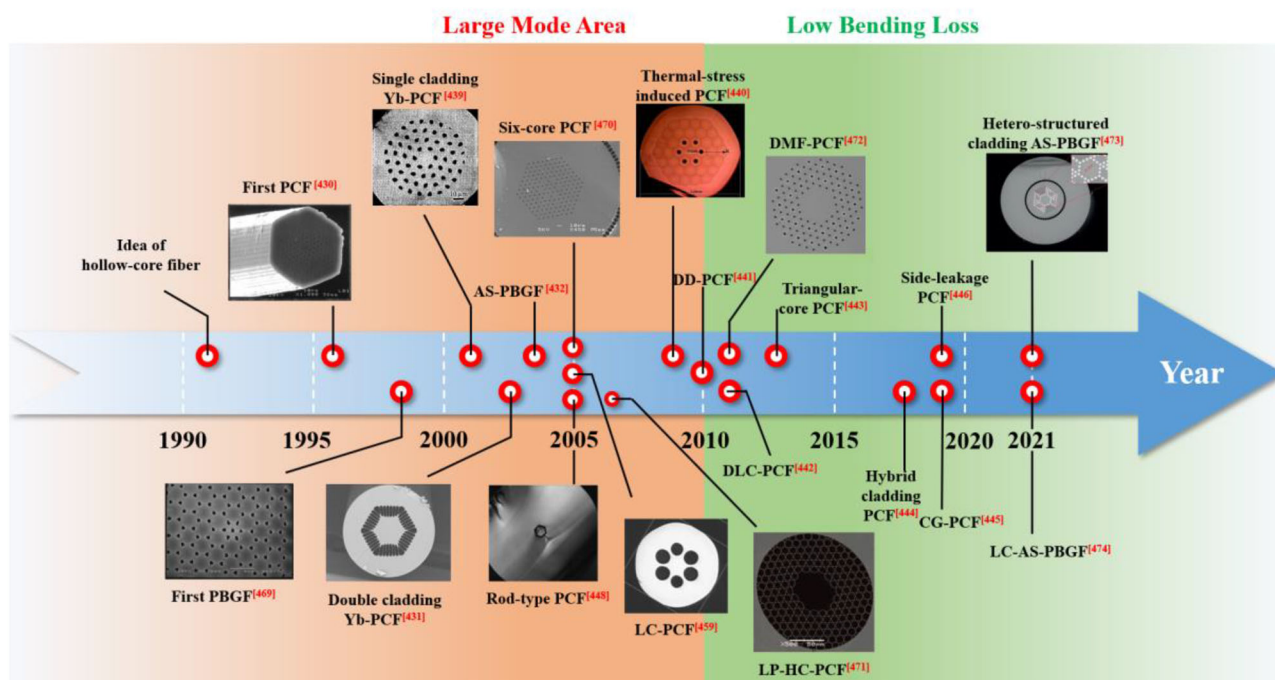


Figure 13. Chronological overview of the leading development of the LMA-PCF.^[468] First reported PCF. Reproduced with permission.^[430] Copyright 1996, Optical Society of America. Photonic bandgap fiber (PBGF). Reproduced with permission.^[469] Copyright 1998, AAAS. Single cladding Yb-doped PCF. Reproduced with permission.^[439] Copyright 2001, IEEE. Double cladding Yb-doped PCF. Reproduced with permission.^[431] Copyright 2003, Optical Society of America. All-solid photonic bandgap fiber (AS-PBGF).^[432] Six-core PCF. Reproduced with permission.^[470] Copyright 2005, Optical Society of America. Leakage channel PCF (LC-PCF). Reproduced with permission.^[459] Copyright 2005, Optical Society of America. Rod-type PCF. Reproduced with permission.^[448] Copyright 2005, Optical Society of America. Large pitch hollow-core PCF (LP-HC-PCF). Reproduced with permission.^[471] Copyright 2006, Optical Society of America. Thermal stress-induced large mode area PCF. Reproduced with permission.^[440] Copyright 2009, Optical Society of America. Double air-hole diameter PCF (DD-PCF).^[441] Distributed mode filtering PCF (DMF-PCF). Reproduced with permission.^[472] Copyright 2011, IEEE. Double lattice constant PCF (DLC-PCF).^[442] Triangular-core PCF.^[443] Hybrid cladding PCF.^[444] Side-leakage PCF.^[446] Circular gradient-diameter PCF (CG-PCF).^[445] Hetero-structured cladding all-solid photonic bandgap fiber (AS-PBGF). Reproduced with permission.^[473] Copyright 2021, IEEE. Leakage channels enabled multi-resonant AS-PBGF (LC-AS-PBGF).^[474]

sol-gel technique,^[437,438] various PCF structures have been emerging consecutively. As shown in **Figures 13** and **14** the early studies focused on increasing the effective MFD and effective field area (EFA) of the fibers, achieving significant progress in EFA from 176 to over 10 000 μm^2 .^[77,439,440] Unfortunately, very large mode area PCFs (VLMA-PCFs) with MFD > 50 μm are almost rod fibers, experiencing deadly bending loss that slows down the development of compact fiber laser systems. Therefore, many new flexible designs include double air-hole

diameter,^[441] double lattice constant,^[442] triangular core,^[443] hybrid cladding,^[444] circular gradient diameter,^[445] etc., have been proposed in the past decade. Simulation results prove that these fibers can maintain a bending loss of less than 10^{-2} dB m^{-1} in the fundamental mode^[368,441,442,445,446] and EFA exceeding 2000 μm^2 ^[446] at a bending radius close to 10 cm. Nonetheless, the increased difficulty and complexity of fabrication caused by noncircular fiber symmetry and additional codoping to eliminate index mismatching delay their practical applications.^[368] The

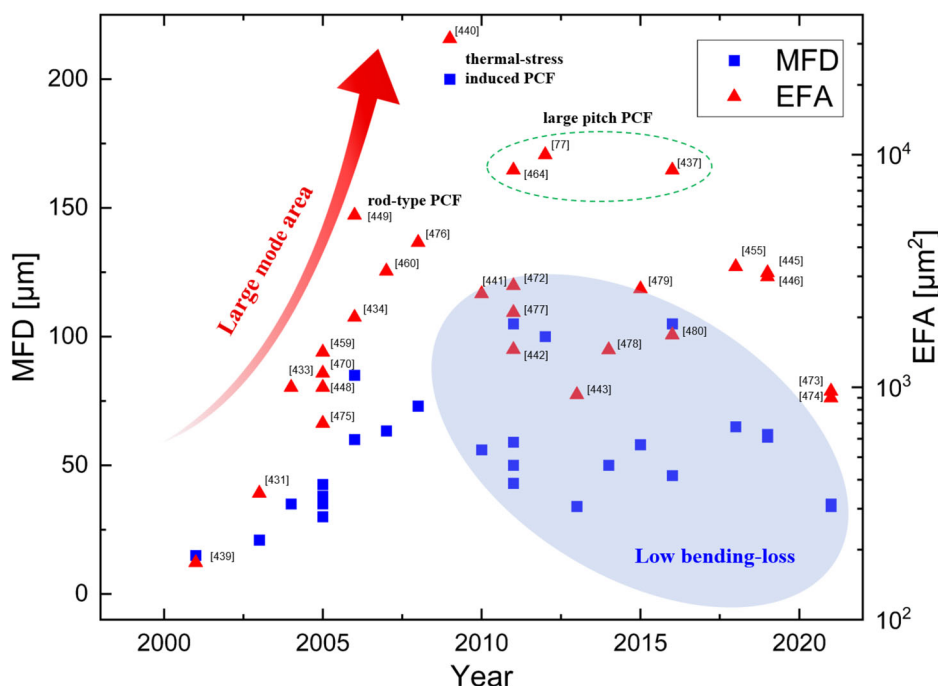


Figure 14. Changes in PCF mode field diameter (MFD) and effective field area (EFA) over the last 20 years. (The references here correspond to the EFA).^[77,431,433,434,437,439–443,445,446,448,449,455,459,460,464,470,472–480]

classic fibers used in high-power lasers are mainly rod-type PCFs, leakage channel PCFs (LC-PCFs), and large pitch PCFs (LPFs). Their characteristics and functions will be further described next.

Small air holes with low d/Λ can easily achieve robust SM propagation;^[433] however since the NA has to be decreased with the increase of core size, the scaling of the PCF is generally limited by macrobending loss and microdeformation loss caused by mode mixing.^[447] The rod-type design, meaning an outer diameter of a few millimeters and a longitudinal length of several tens of centimeters, is an effective way to obtain a large core and negligible bending loss.^[448] Two special waveguides in the double-clad structure make low-brightness pumping possible,^[429] and the larger outer diameter guarantees the high mechanical stability and heat dissipation capability of PCFs,^[448] which enables enormous performance improvement of fiber laser systems, particularly for the pulsed amplifiers. Nanosecond fiber lasers with a few hundreds of kilowatt average power and multimillijoule pulse energy have been generated from rod-type PCFs with core sizes in the range of 60 to 100 μm .^[434,449–452] Furthermore, these fibers are also the basis of 100 MW-level to GW-level peak power fs fiber amplifiers, which are routinely operated in state-of-the-art fiber-CPA systems.^[453–457] Nevertheless, the challenging fabrication of minute air hole arrays reduces the reproduction of these VLMA fibers.^[77] By further increasing the core diameter, rod PCFs are not SM according to strict analysis.^[458]

Effective fundamental mode propagation is also achievable in few-mode fibers by using applicable fiber designs that offer different modal propagation losses or excitation and amplification conditions.^[458] LC-PCFs comprising one or several rings of very large air holes provide much higher leakage losses for HOMs while confining the FM in the core by the high-contrast

air–glass boundary.^[459] A Yb-doped double-clad SM-LCF with a mode diameter of 63.4 μm and a mode area of 3160 μm^2 has been reported.^[460] Although this fiber displays good bending resistance, mode shrinking caused by coiling has to be considered for VLMA fibers, which can become the most limiting factor for EFA scaling.^[461] HOMs delocalization, i.e., poor overlap with the core is a superior novel approach to achieve effective mode discrimination in a VLMA rod-type fiber. LPFs with pitch Λ larger than 10 times the wavelength λ is the first representative of this idea.^[77] Inside the relatively open structure, HOMs are not easily excited by a Gaussian beam, and their amplification factor is substantially reduced; thus, only FM propagates with high gain.^[458] A SM-LPF structure with a record large core diameter of 135 μm and a mode field area over 10 000 μm^2 has been proposed and practically applied in fiber amplifiers.^[462] In addition, the simple nonresonant structure enables LPFs to employ an index-depressed core to precompensate for the thermal index profile in high power operation.^[371] With outstanding scalability and robustness, LPFs currently hold the majority of the representative results for high-performance pulsed fiber laser systems. Typical examples are the 26 mJ pulse energy^[462] and 2 kW average power^[463] directly obtained from the LPF-based MOPA system. In 2011, Eidam et al. designed a VLMA-LPF with MFD $\approx$ 105 μm and obtained 480 fs diffraction-limited compressed pulses at 3.8 GW peak power from the monolithic CPA configuration.^[464] The record was not broken until 2017 when Lavenu et al. employed nonlinear compression in a Xenon-filled hollow capillary after LPF-based CPA system to generate 14 fs ultrashort pulses with 5.8 GW peak power.^[454]

Additionally, a longer emission wavelength also benefits mode area scaling,^[458] therefore, Tm-doped PCFs working at approximately 2 μm allow for an MFD twice as large as Yb-doped

PCFs under the same NA and cutoff frequency value. Moreover, Tm-doped fiber amplifiers have a much stronger thermal bearing capability and higher self-focusing threshold.^[465,466] They have excellent power scaling prospects and have already generated fs pulses with 6.9 GW peak power^[47] and 1 kW average power.^[467]

3.5.2. CBC by PCFs

Although VLMA-PCFs have shown great significance in the power scaling of laser systems, the performance of monolithic PCF-based amplifiers is generally limited by material damage.^[481] One way to avoid this problem is coherently combining fiber amplifiers to deliver peak/average powers many times higher than those achievable with a single amplifier. Klenke et al. reported an ultrafast laser system comprising four parallel Yb-doped LPF amplifiers producing a compressed pulse energy of 5.7 mJ. The total combined pulse peak power was 22 GW, which could be boosted to 30 GW in the case of a completely flat phase.^[481] In 2021, they updated the system to deliver 120 fs diffraction-limited pulses with 10 mJ pulse energy. The peak power was boosted to 68 GW while 1 kW average power was obtained.^[482] Note that divided-pulse amplification (DPA), which uses birefringent crystals or delay lines to split and recombine pulses, is a promising technology to accelerate the improvement of PCF amplifiers.^[483] Kienel et al. illustrated a DPA system that amplified four temporally divided pulses and output fs pulses with pulse energy up to 12 mJ after compression, resulting in peak power of 45.8 GW.^[484] Later, Stark et al. utilized two electro-optical modulators to control pulse division and realized a compressed pulse energy as high as 23 mJ. A total of 96 pulses are combined spatially and temporally generating pulses with a peak power of nearly 100 GW.^[485] Table 3 lists the parameters of some representative PCF-based fs fiber systems with monoamplifiers or combined amplifiers in the past 5 years.

Apart from the large mode area and endless SM properties, purposeful manipulation of fiber structure endows PCFs with enhanced nonlinearity, modified dispersion, high birefringence, and other unique optical characteristics that facilitate their application to cover not only high power laser but also many other disparate fields such as nonlinear optics,^[216] interferometry,^[496] beam deliver,^[497] sensing,^[498,499] microscopy,^[500] etc. For example, gas-filled hollow-core PCFs (HC-PCF) with flexible pressure tunability of the zero-dispersion wavelength (ZDW) allows the generation of supercontinuum from the deep- or vacuum-ultraviolet region^[217,219] to mid-infrared.^[501] Recently, supercontinuum generation has extended down to a record 124 nm^[217] and up to 4 μm ^[502] based on hydrogen-filled kagomé-style HC-PCF and argon-filled anti-resonant HC-PCF, respectively. Due to the lower transmission loss and better dispersion control than conventional capillaries, gas-filled HC-PCFs are also the superior alternative for spectral broadening and pulse compression that can produce 3.8 fs pulses with peak power higher than 0.6 GW.^[503] In addition, hydrogen-filled HC-PCFs have the advantages of narrow linewidth, high efficiency, and tunability, which provide a new method for 1.7 μm Raman lasers.^[504] Unfortunately, unique structures bring so many competitive advantages and the problem of cleaving and splicing of fibers that light

Table 3. The parameters of the fs fiber amplification system based on PCF over the last 5 years. Peak power (P_{peak}), average power (P_{avg}), pulse energy (E_p), pulse duration (τ_p).

Year ^[Ref]	Fiber type	MFD [μm]	P_{avg} [W]	P_{peak} [GW]	E_p [μJ]	τ_p [fs]	M^2
Chirp-pulse amplifier							
2016 ^[453]	Yb-LMA-PCF	35	100	0.37	100	270	–
2017 ^[454]	Yb-DMF-PCF	60	24	5.8	120	14	<1.3
2018 ^[455]	Yb-rod-PCF	65	32	0.209	32	153	1.3
2019 ^[486]	Yb-LMA-PCF	35	13.5	0.113	27	172	1.2
2019 ^[456]	Yb-rod-PCF	65	80	1.21	400	330	<1.2
2020 ^[457]	Yb-rod-PCF	65	33.2	0.244	66.4	272	1.1
2020 ^[487]	Yb-rod-PCF	65	50	0.12	100	830	1.3
2016 ^[488]	Tm-LPF	63	28.7	1.96	470	200	–
2017 ^[47]	Tm-LPF	63	6.82	6.9	620	90	–
2018 ^[489]	Tm-LPF	35	790	–	–	250	<1.15
2018 ^[467]	Tm-LPF	35	1060	0.05	13.2	265	<1.1
2020 ^[490]	Tm-LPF	63	108	0.73	260	250	–
Coherent beam combination							
2016 ^[491]	Yb-LPF	64	874	12	3.3 mJ	275	1.1
2016 ^[484]	Yb-LPF	–	700	45.8	12 mJ	260	<1.1
2018 ^[492]	Yb-LPF	63	1830	9.58	2.3 mJ	240	1.8
2019 ^[485]	Yb-LPF	–	674	92	2.3 mJ	235	–
2020 ^[493]	Yb-rod-PCF	65	1100	–	–	300	–
2021 ^[494]	Yb-rod-PCF	65	–	2.3	0.12 mJ	44	–
2021 ^[482]	Yb-LPF	62	1000	68	10 mJ	120	<1.2
2020 ^[495]	Tm-LPF	35	>114	>1.3	>0.23 mJ	<120	<1.2

enters PCFs almost through spatial coupling. With the continuous development of laser technology and the progress of fiber fabrication, compact all-fiber PCF-based laser systems will have greater application prospects in the future.

4. Summary and Outlook

Compared to traditional rod-type, zigzag-type, or active-mirror slabs, the peak power of ≈ 10 ns pulse can be scaled from sub-GW to ≈ 10 GW by a multi-slab structure expand the allowed size of the slabs due to gas colling. A record pulse energy of 250 J has been reported and the further breakthrough is limited by the power of the pulsed pump and crystal aperture.^[105] Meanwhile, a CPA-free ps pulse with several kW average power and $\approx$ GW peak power can be obtained from a multipass disk amplifier due to excellent heat dissipation performance. Further realizing the power output of 10 000 W is limited by the power of the zero-phonon-line pumping.^[505] Furthermore, the energy of a compressed ≈ 1 ps pulse can go beyond 100 mJ (<1 J) by a hybrid disk amplifier which experiences a repetition reduction by a per-regenerative amplification and an energy boost by several master multi-pass disk amplifiers. The master thin-disk can be replaced by a thick-disk or even an active-mirror slab, and over 1 J energy have been achieved.^[64] This hybrid amplification scheme will be the preferred strategy for realizing compact commercial TW lasers in the near future.

OPCPA can realize a peak power of TW and even several PW with large diameter BBO, LBO.^[56,85] At present, the CPA technology based on Ti:sapphire has achieved a peak output greater than 10 PW. As the crystal diameter has reached more than 200 mm, advanced pulse pumping methods and parasitic oscillation suppression methods are necessary.^[140,506] The further increase of the peak power is limited by the strength of the pulse pump, the damage of the compression grating, and the growth of a larger crystal aperture. In the future, driven by relativistic attosecond science, the peak-power of a few-cycle fs pulse with high temporal contrast need to exceed PW. Small size crystals in the TW scenario will be insufficient for PW and the contrast deterioration caused by large aperture crystals will become a new challenge.

The nonlinear effect is the main limiting factor of high strength fiber amplifiers; as for SRS suppression, although some spectrally selective fibers are still not currently used in high power amplifiers, it has reference value for future work. Another chirped fiber grating has direct Raman filtering capabilities under kW-level average power.^[80] In the future, breakthroughs in manufacturing larger core diameters by fs-laser direct etching technology will be a powerful way to achieve higher-intensity filter gratings. It is conceivable that this direct multiphoton etching process without high-pressure hydrogen treatment is promising to directly etch multiple integrated filter gratings on the gain fiber, which can not only filter out the residual Raman light but also achieve ASE suppression. This will promote the design and realization of a more flexible, compact and stable fiber laser system. Of course, with the further increase of power, the damage of the fiber coating caused by the leaked-light will be a problem to be faced in the future. Higher damage-resistant coating materials may be more pressingly needed. On the other hand, the inelastic SBS, which limits the power of narrow-linewidth fiber laser can be effectively eliminated by appropriately extending the line width by phase modulation. But this is at the cost of sacrificing the coherence of the laser, and how to compensate for the broadened bandwidth will be a concern. In addition, it would be a more reasonable solution to develop a new type of optical fiber to realize the gain separation of light field and sound field. This rises new challenges for the design of optical fiber waveguide material.

Using larger core diameter fiber is a direct means to suppress nonlinearity, but this will introduce HOMs and trigger the TMI effect. The mechanism of this thermally induced instability phenomenon and the suppression methods have been extensively studied, but there is no particularly outstanding suppression result. An exciting process called “beam cleaning” opens the door to improve the beam quality and will cause extensive research.^[363]

To balance the contradiction between nonlinear suppression and TMI, some LMA functional fibers have been designed. SM purity has been strictly verified, and they also show outstanding nonlinear suppression in practical applications. The widely used CCCFs and TDCFs have been realized several kW TMI-free CW output and MW-level peak power output with CPA free. These performances are almost comparable to PCF. However, the PCF still maintains the current peak record of the optical fiber. Moreover, the current coherent beam combining system of the PCF amplifier has achieved a peak output of 68 GW.^[482]

As discussed above, it is clear that the requirements for the gain medium to achieve high peak and average power are contradictory. Although the peak power of solid-state lasers has reached

PW level, limited by thermal lens and stress birefringence effects of the ultra-large diameter crystals, the average power is lower than 100 W. However, besides the peak of PW an average power of MW is also necessary for a practical laser-based particle accelerators. Benefit from the high efficiency and compactness of a single fiber laser channel, this ambitious goal will be accomplished by spatial coherent combination and temporal stacking.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

disks, fiber lasers, high-power lasers, nonlinear suppression, slabs, ultra-fast pulses

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